

# **Investigation of Lithosphere-Atmosphere-Ionosphere Coupling associated with Earthquakes in Pakistan from Machine Learning Procedures**

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# **Investigation of Lithosphere-Atmosphere-Ionosphere Coupling associated with Earthquakes in Pakistan from Machine Learning Procedures**

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Institute of Space Technology  
in partial fulfillment of the requirements  
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by

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**Amna Hafeez**



## ABSTRACT

Earthquake causes human and material losses, every year. However, a reliable precursor for an earthquake is inevitable to relieve humans from its fatality. In this work, a multidisciplinary overview of short-term earthquake precursory responses is conducted in the atmosphere and ionosphere. For this purpose, statistical and machine learning procedures are implemented on Total Electron Content (TEC) from Global Positioning System (GPS) and Land Surface Temperature (LST) from Moderate Resolution Imaging Spectroradiometer (MODIS) to detect possible pre-and post-seismic perturbations within the preparation period of large magnitude earthquakes. The main findings are summarized as below.

- This study shows abnormal co-located atmospheric attribute of the  $M_w$  5.4 Mirpur event from LST over the epicenter during 5-10 days before/after the main shock day. The abnormal variations for this earthquake are also validated by GPS TEC retrieved from Sukkur (27.8° N, 68.9° E) station. The anomalous pattern is duly verified by statistical and machine learning procedures within the observed window of 5-10 days before/after the main shock. The same techniques of this earthquake are also implemented on  $M_w$  7.7 Awaran earthquake to aid in the development of lithosphere atmosphere ionosphere coupling (LIAC) hypothesis.
- LST and TEC anomalies can be clearly seen as possible precursors before the  $M_w$  7.7 Awaran earthquake and responses after the main shock. Every implemented technique indicates abnormal LST and TEC 5-7 days before seismic breeding day and subsequent rise in temperature during 2 days after the main shock.
- These results suggest the huge emanation of earthquake energy before/after the main shock. With the current satellite clusters, it is impossible to confidently forecast earthquakes. However, the extensive analysis can be extended to more satellite and ground-based measurements in Pakistan and other countries to reveal the pattern of air ionization over the epicenter to atmospheric and ultimately ionosphere.

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## **LIST OF ABBREVIATIONS**

|         |                                                |
|---------|------------------------------------------------|
| SATNAV  | Satellite Navigation                           |
| GNSS    | Global Navigation Satellite Systems            |
| GPS     | Global Positioning System                      |
| MEO     | Medium Earth Orbit                             |
| GLONASS | Global'naya Navigatsionnaya Sputnikova Sistema |
| EU      | European Union                                 |
| BDS     | BeiDou navigation satellite system             |
| GEO     | Geostationary Earth Orbit                      |
| IGSO    | Inclined Geosynchronous Orbit                  |
| SBAS    | Satellite based augmentation system            |
| SAR     | Search and Rescue                              |
| WGS     | World Geodetic System                          |
| EQ      | Earthquake                                     |
| TEC     | Total Electron Content                         |
| LST     | Land Surface Temperature                       |
| LAIC    | Lithosphere Atmosphere Ionosphere Coupling     |
| USGS    | United States Geological Survey                |
| STEC    | Slant TEC                                      |
| VTEC    | Vertical TEC                                   |
| IQR     | Inter-quartile Range                           |
| CWT     | Continuous Wavelet Transformation              |
| UV      | Ultraviolet                                    |
| EM      | Electromagnetic                                |
| EUV     | Extreme Ultraviolet                            |
| HF      | High Frequency                                 |
| UB      | Upper Bound                                    |
| TECU    | Total Electron Content Unit                    |
| LB      | Lower Bound                                    |
| NN      | Neural Network                                 |
| MLP     | Multi-Layer Perceptron                         |
| NARX    | Nonlinear AutoRegressive eXogenous             |
| MODIS   | Moderate Resolution Imaging Spectroradiometer  |

|      |                                               |
|------|-----------------------------------------------|
| UT   | Universal Time                                |
| ML   | Machine Learning                              |
| URSI | International Union of Radio Science          |
| TIR  | Thermal Infrared Radiation                    |
| ECSS | European Cooperation on Space Standardization |
| hmF2 | F2-peak height                                |
| SIAs | Seismic Ionospheric Anomalies                 |
| NmF2 | F2 layer peak electron density                |
| STEC | Slant Total Electron Content                  |
| LT   | Local Time                                    |



## 1 Introduction to Earthquake precursors in Pakistan

Pakistan is located geographically between Latitude ( $60^{\circ}\text{E}$ - $78^{\circ}\text{E}$ ), and Longitude extension ( $24^{\circ}\text{N}$ - $37^{\circ}\text{N}$ ). As one of Asia's most seismically active areas, it has a high density of active faults. The area has many active faults and is near the intersection of three plate boundaries. Northern and Western parts are particularly vulnerable to earthquakes. As Pakistan is one of the prone zone, it has a vast history of major devastating events that led to many demolitions and devastations. Earthquakes, like other natural disasters, are unpredictable and one of the most deadly. The inability of science and technology to anticipate the exact timing or location of this massive tragedy is alarming, but it is unavoidable for mankind. A list of natural disasters has claimed the lives of 6037 people between 1993 and 2002, with another 8.9 million people also affected (World Disasters Report, 2003). The earthquake of 8 October 2005 killed  $> 80\,000$  people and caused 3.5 million homes to be destroyed.

Earthquakes affect humans in 7 ways, there are Loss of lives, Social Impact, Burden due to Injuries, Psychological Trauma, Economic damage, Environmental Damage (Table 1.1).

**Table. 1.1.** Major Pakistan's Earthquakes and their respective destructions ([www.dawn.com](http://www.dawn.com))

| Year | Affected Area                 | Magnitude | Causalities | Infrastructure damage |
|------|-------------------------------|-----------|-------------|-----------------------|
| 2005 | Kashmir and Northern areas    | 7.6       | 80,000      | $>32,335$ houses      |
| 2011 | Southwest Pakistan            | 7.2       | 3           | 200 houses            |
| 2013 | Quetta                        | 7.9       | 34          | 10,000 houses         |
| 2016 | Khyber Pakhtunkhwa and Punjab | 7.1       | 6           | $>20$ houses          |

A ground-based receiver at a known position that measures pseudorange and phase towards all

GNSS satellites in view provides useful measurements for estimating the slip moment along the active fault regions. In recent years, high-speed GNSS tracking networks have been established, allowing earthquakes to monitor. As GNSS signals are becoming more and more available, innovative uses for GNSS signals exploring Earth's atmosphere, ionosphere, and surface will continue to flourish. In the future, this will enable us to measure and predict environmental changes on Earth or the upper atmosphere around the time of an earthquake (occurrence, before, during, after) with increasing accuracy.

## **Objective and Motivation**

Due to the fact that GNSS signals provide insights over the ionospheric conditions by calculating the relative amount of TEC content in the ionosphere. The researchers have suggested that TEC concentrations in the ionospheric may also be affected by earthquakes, a discovery that will contribute to a better understanding of EQ-ionosphere interactions[1]. The fundamental purpose of this research is to show the synchronization of atmospheric and ionospheric parameters in time and space, as well as to demonstrate the precursory parameters with multiple approaches. Analysis of seismically produced disturbances at the ground level and in the ionosphere as probable precursors to earthquakes of magnitude  $>5.4$  is presented as an example of our technique.

In addition to Ionospheric TEC analyses, the monitoring of atmosphere anomalies from space-borne atmospheric sensors may shed light on the predictive nature of future earthquakes (EQs). In this research, the possible relationship between the EQs and the ionosphere-and-atmospheric variations for the EQs of  $M_w > 5.4$  during 2013-2019. The GPS-TEC datasets will be retrieved from the Sukkur GPS station. The anomalies will be quantified using a combination of statistical, classical, and intelligent methods. All the analyses and techniques will be performed for the different earthquakes in Pakistani region.

## **Content of the Study**

This study is organized as follows:

**Chapter 2** gives the evolution of GNSS which includes Introduction to GNSS, present and future systems, GNSS for Geodynamics and lastly seismic precursors.

**Chapter 3** comprises of background and literature review

**Chapter 4** includes proposed methodology to carry out this research work

**Chapter 5** gives detailed discussion on the significance of work and highlights necessary results  
**Chapter 6** has conclusions and recommendations

## **2 INTRODUCTION**

### **2.1 Navigation**

The positioning of points on the surface of the Earth has been determined for hundreds of years by observations of distant objects. Reflecting mirrors on mountain tops were replaced by high-altitude flares and rockets. The skill of celestial navigation has been developed for centuries through the observation of the Sun, stars, planets, and other celestial objects. However, it was not until the dawn of the space age that an accurate system for locating and navigating became possible on a global scale

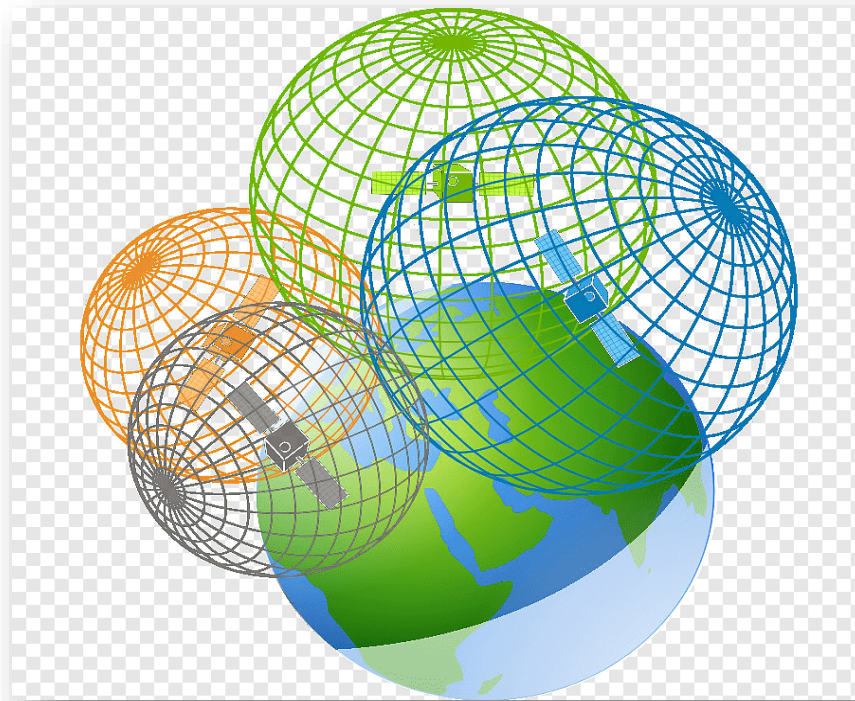
#### **2.1.1 Satellite based Positioning**

As a satellite-based positioning system, artificial satellites are used to determine the locations of observing sites located in the air, ground, and sea. Due to this, satellite positioning should not be confused with its often used abbreviation. During operational satellite-based positioning, satellite positions are assumed to be known at each epoch. The orbit of the satellite may be determined by evaluating the Doppler shift of 20 MHz radio signals that the satellite transmits at a particular point. In due course, researchers discovered that if a satellite's orbit could be identified, a receiver's position could be derived from its shift. Global navigation satellite system (GNSS) refers to both individual satellite-based positioning systems and their combinations or augmentations [2].

#### **2.1.2 Principle of Satellite Positioning**

The use of operational satellites enables the user to determine the position using data such as latitude, longitude, and height. In this procedure, ranges or disparities measured by satellites are used to perform a simple resection technique. Using this method, only three satellite ranges are required because the intersection of three spheres produces three unknowns e.g., latitude, longitude, and height that can be computed by using the three range equations. And with the help

of fourth satellite time difference can be calculated as well (Fig. 2.1).



**Fig. 2.1.** Positioning principle using four GNSS Satellites

**Source:** <https://gisgeography.com/trilateration-triangulation-gps/>

## 2.2 GNSS Systems

In order to address the shortcomings of earlier reference systems (NAVSTAR), the US military created the GPS (Global Positioning System). GPS is more accurate, inexpensive, and quickly able to answer queries such as "where am I, what time is it, how fast is it?" anywhere on the globe, at any time.

Military officials in Russia use the Globalnaya Navigazionnaya Sputnikovaya Sistema (GLONASS), which is the Russian equivalent of GPS. GLONASS differs from GPS in three aspects: the control segment, the satellite, and the signal structure [3].

The Galileo system is part of Europe's contribution to the future GNSS. Since 1999, Galileo has become the name of the European contribution to satellite navigation, which has since become synonymous with the European GNSS.

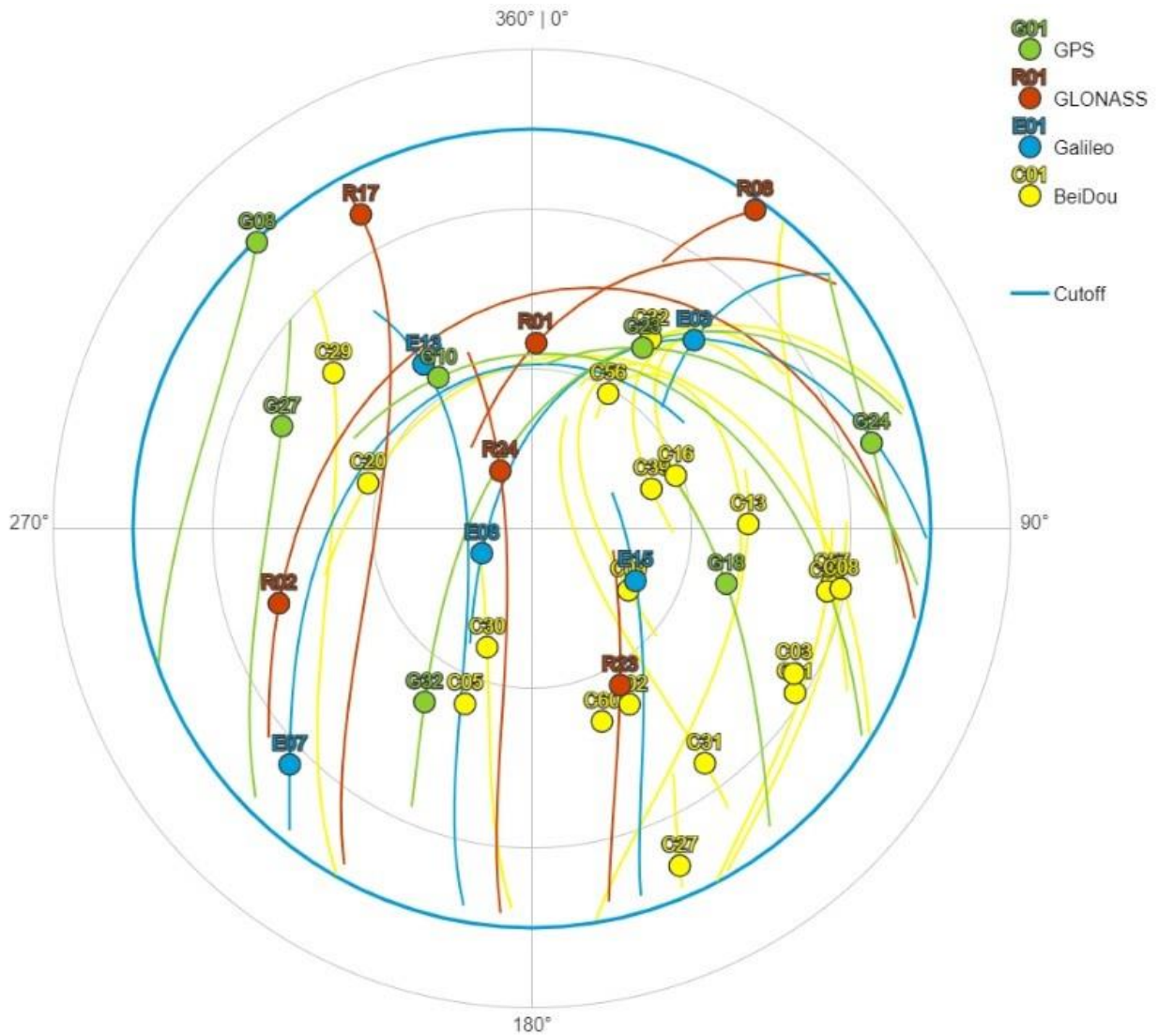
Developed from the first-generation regional system, Compass is a Chinese system. Satellite

navigation was first explored in the early 1980s, using the same regional system concept proposed and developed by the GEOSTAR Corporation. A two-way range method will be used to determine position by means of two geostationary satellites.

The basic information of different individual satellite system in GNSS is summarized in table 2.1.

**Table. 2.1.** GNSS satellite systems with basic briefing

| System                   | GPS<br> | GLONASS<br> | BeiDou<br> | Galileo<br> |
|--------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Orbit</b>             | MEO                                                                                      | MEO                                                                                          | MEO, IGSO, GEO                                                                               | MEO                                                                                            |
| <b>Satellite numbers</b> | 24                                                                                       | 24                                                                                           | 27,3,5                                                                                       | 30                                                                                             |
| <b>Constellation</b>     | 6 planes<br>56° inclination                                                              | Walker(24/3/1)<br>64.8° inclination                                                          | Walker(24/3/1)<br>55° inclination                                                            | Walker (24/3/1)<br>56° inclination                                                             |
| <b>Services</b>          | SPS, PPS                                                                                 | SPS, PPS                                                                                     | OS, AS, WADS, SMS                                                                            | OS, CS, PRS                                                                                    |
| <b>Initial Services</b>  | 1993                                                                                     | 1993                                                                                         | 2012                                                                                         | 2016                                                                                           |
| <b>Origin</b>            | USA                                                                                      | Russia                                                                                       | China                                                                                        | Europe                                                                                         |
| <b>Coverage</b>          | Global                                                                                   | Global                                                                                       | Global                                                                                       | Global                                                                                         |
| <b>Frequency (MHz)</b>   | L1 1575.42<br>L2 1227.60<br>L5 1176.45                                                   | L1 1602.00<br>L2 1246.00<br>L3 1202.025                                                      | B1 1561.098<br>B2 1207.14<br>B3 1268.52                                                      | E1 1575.42<br>E5a 1176.46<br>E5b 1207.14<br>E6 1278.75                                         |



**Fig. 2.2.** Satellite based configuration (sky plot) at GNSS ground station in a particular epoch

**Courtesy:** <https://www.gnssplanning.com/>

In addition to other techniques, Space Based Augmentation Systems (SBAS) and Ground Based Augmentation Systems (GBAS). SBAS includes the US Wide-Area Augmentation System (WAAS), the European Geostationary Navigation Overlay Service (EGNOS), and the Japanese Multifunctional Transport Satellite (MTSAT) Space-Based Augmentation System (MSAS). As an extension to existing Medium Earth Orbit (MEO) satellite constellations, these systems add Geostationary (GEO) or geosynchronous satellites (Fig. 2.2).

## 2.3 GNSS Segments

Every GNSS system consist of three parts including constellation configuration, tracking and monitoring stations and user segment. All these segment are almost same in all technologies which are altered together as GNSS ([Table. 2.2](#)).

**Table. 2.2.** Information of different GNSS segments

| Segments              |                                                                                   | GPS                                                                                  | GLONASS                                                                   | BeiDou                                                                      | Galileo                                                                           | Working                                                                                                                                                                                        |
|-----------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Space Segment</b>  |  | 30 operational satellites                                                            | 10 operational satellites                                                 | 35 operational satellites                                                   | 10 operational satellites                                                         | Transmit ranging signals and data messages                                                                                                                                                     |
| <b>Ground Segment</b> |  | 16 monitoring stations along with Master control station located at Colorado Springs | 2 monitoring stations along with Master control station located at Prague | 32 monitoring stations along with Master control station located at Beijing | 4 monitoring stations along with Master control station located at Krasnoznamensk | <ul style="list-style-type: none"> <li>Tracks and maintains the satellites in space</li> <li>Monitors satellite health and signal integrity and maintains the orbital configuration</li> </ul> |

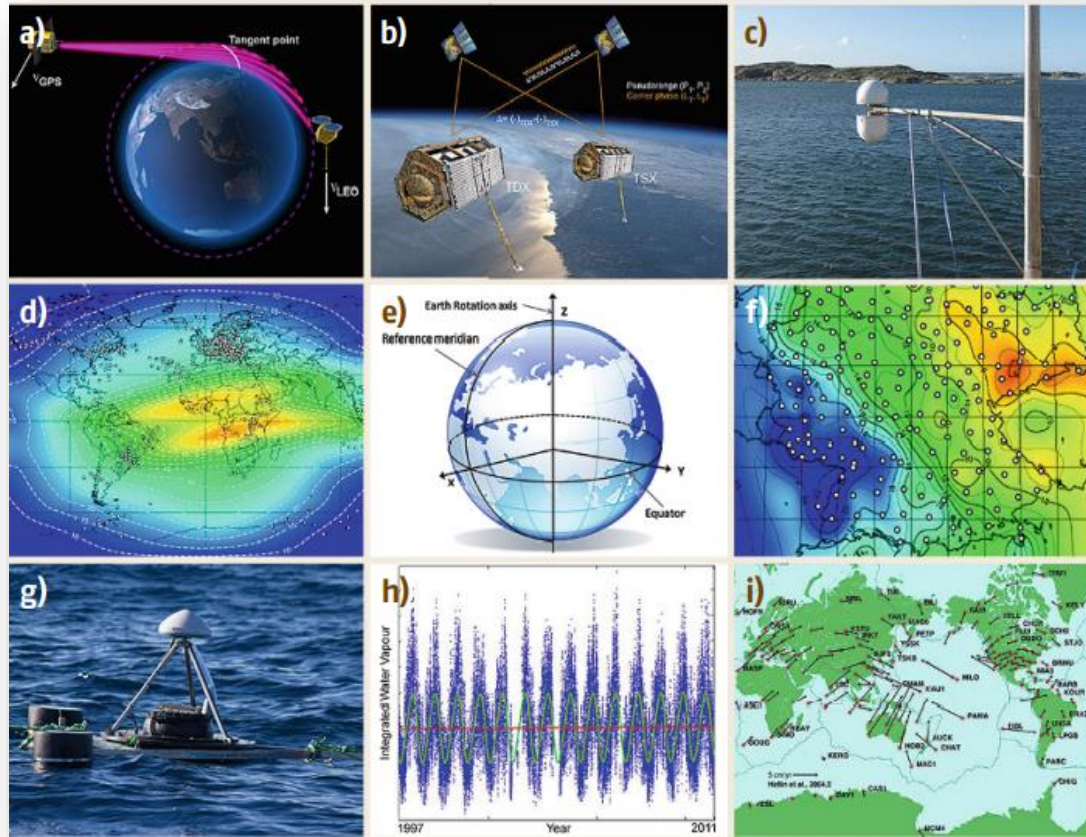
|                         |                                                                                     |                               |                               |                               |                            |                                                                                                                                                                                                                                                                        |
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|                         |                                                                                     |                               |                               |                               |                            | of the<br>satellites<br><ul style="list-style-type: none"> <li>• Updates the<br/> satellite<br/> clock<br/> corrections<br/> and<br/> ephemerides<br/> as well as<br/> numerous<br/> other<br/> parameters<br/> essential to<br/> determining<br/> user PVT</li> </ul> |
| <b>User<br/>Segment</b> |  | User<br>receiver<br>equipment | User<br>receiver<br>equipment | User<br>receiver<br>equipment | User receiver<br>equipment | Performs the<br>navigation, timing,<br>or other related<br>functions                                                                                                                                                                                                   |

## 2.4 Earth Science Applications of GNSS

Earth science and environmental system investigations utilizing GNSS remote sensing are among the growing applications of GNSS technology ([Fig. 2.3](#)). Geophysical survey dual frequency



receivers operate continuously in the vicinity of geophysical significant locations. In terms of measuring the long-term movement of tectonic plates, GNSS sounding has numerous applications [4].



**Fig. 2.3.** Examples of scientific GNSS applications; from atmospheric sensing and reference frame studies to tectonic monitoring (Photos courtesy of UCAR 2007 (a); P. Kuss, DLR, NASA (b); J. Löfgren (c); N. Jakowski, DLR (d); J. Legrand & C. Bruyninx, ROB (e); G. Dick, GFZ (f); IMOS (g); G. Elgered, Chalmers (h); NASA, JPL-Caltech (i))

## 2.5 GNSS for Geodynamics

Around a century before GNSS, geodetic position was used to analyze tectonic motion, and the first key models of tectonic geodesy were developed during this period [5]. A pioneering study initially examined displacements caused by large earthquakes, which was the only tectonic

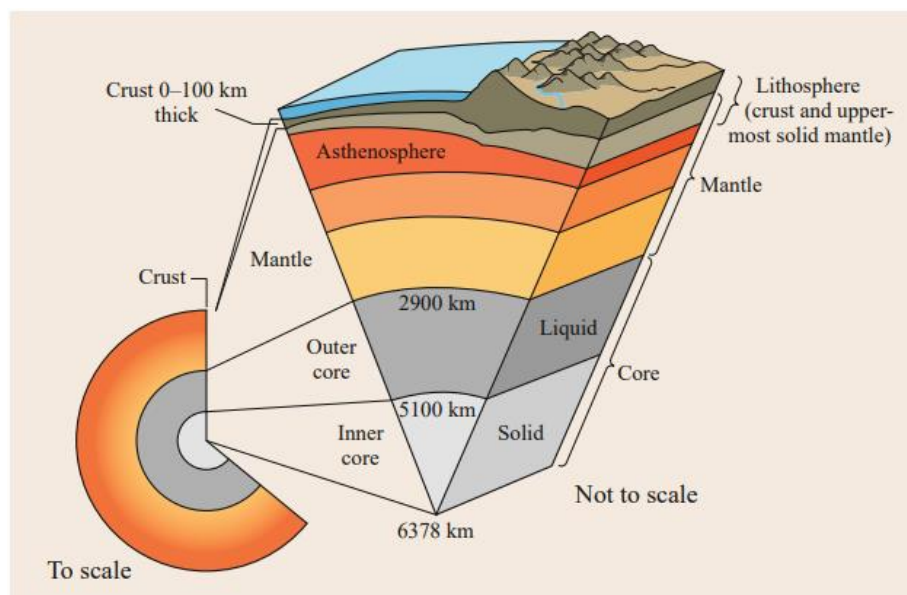
movement large enough to be seen at the time. A study i.e., elastic rebound hypothesis [6] examined displacements before and after the 1906 San Francisco earthquake, which supported Gilbert's theory that earthquakes resulted from a sudden release of strain built up over an extended period [3]. The first GNSS experiments were carried out in the 1980s, when modern geodesy, while still proving its full capabilities, had yet to reach its current level of accuracy. By using existing computational methods, surface deformation can be applied to tectonics, volcanology, and other geodynamic studies. To measure surface deformations intensively on a geographical scale and time scale, a measurement tool was necessary. The first GNSS system, GPS, provided this service.

The following are some of the broad categories of geodynamic studies utilizing GNSS:

- i. Rigid plate motion
- ii. Plate boundary deformation
- iii. Earthquakes and seismology

### 2.5.1 Rigid Plate Motion

The crust, mantle, and core are the three primary chemically different layers that make up the Earth (Fig. 2.4). As for the Earth's mechanical properties, however, these do not correlate with its chemical composition. As a result of the movement of the plates, the lithosphere is formed, incorporating the crust and some of the mantle [5].



**Fig. 2.4.** A depth section showing the chemical layers i.e. crust, mantle, core as well as the

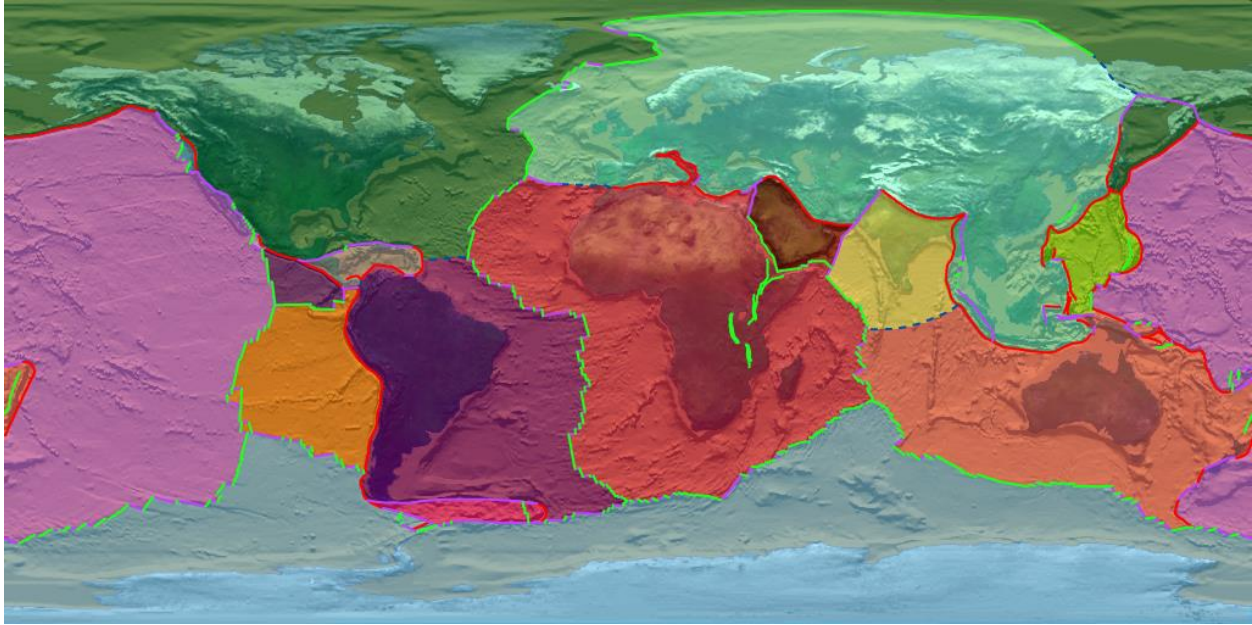
mechanical layering; i.e., asthenosphere, lithosphere, mantle. The lithosphere, which is made up of the crust and the coldest section of the uppermost mantle, includes tectonic plates.

**Source:** United States Geological Survey (USGS)/Wikimedia Commons

Lithospheres behave like elastic substances, while asthenosphere are viscoelastic, meaning they act like fluids over long time periods. The asthenosphere, due to its higher hydration levels, is more vulnerable. There seem to be large plate border zones suggesting the lithosphere has not been entirely elastic, and deformations have accumulated over time. Tectonic zones undergoing active tectonic deformation usually have a lithosphere thickness of less than 50 kms, while stable continental interiors have a lithosphere thickness of more than 100 kms [4].

### **2.5.2 Plate Boundary Deformation and the Earthquake Cycle**

Despite the fact that the basic idea of plate tectonics claimed that all plates were solid and non-deforming, however it was quickly discovered early on that seismicity and active tectonics in the Earth's crust covered a vast area rather than being confined to a narrow band near plate boundaries (Fig. 2.5). Deforming zones along continental crust plate boundaries are called plate boundary zones [7]. Geodesy, tectonics, and earthquakes are related via the concept of earthquake cycles. Models of earthquake cycles describe how constant driving stresses result in a buildup of strain and stress on a fault that can lead to rapid fault slip in an earthquake. There are times when a fault is not moving at all for decades or even centuries, only to suddenly move a great deal in an earthquake. Because of the advent in the field of GNSS, the volume and quality of deformation data has grown in the last few decades, the core models have held up quite well. Since GNSS systems were developed, these models have undergone some refinement, including the discovery of a number of new transient and slow slip incidents, although the basic principles of these models have largely remained unchanged [3].



**Fig. 2.5.** A combined image of the Earth's plates, plate boundaries, and ocean bathymetry

**Source:** NASA

### 2.5.3 Plate Boundary Zones

There is an accumulated deformation in a plate border zone that equals the relative plate motion between two plates. It is equal to the total deformation of a zone that lies overall along a plate boundary between two plates. It is most likely a result of pre-existing lithosphere strength differences, such as differences in rock types, already-existing faults, and so on. A fault slip causes the coefficient of friction of most Earth materials to decrease under the pressure and temperature conditions of the shallow crust, causing an earthquake to accelerate the slip (and further reduce the coefficient of friction). The fault slows and eventually stops moving as soon as the driving stress falls below the frictional force. As stress and strain accumulate over time, subsequent earthquakes result. The earthquake cycle is the result of this relationship [8]. In the earthquake cycle, the inter-seismic phase is the time between earthquakes. After a powerful earthquake, temporary processes occur in the post-seismic phase of the earthquake cycle.

## 2.6 Seismology

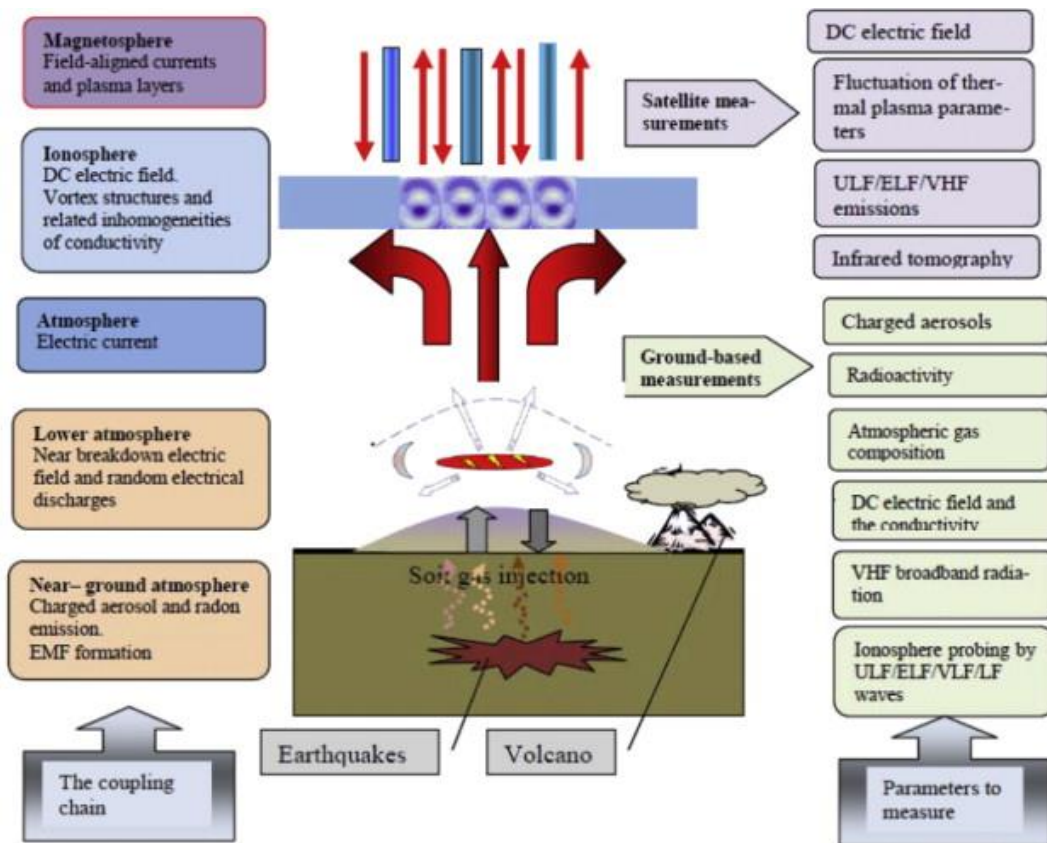
Earthquakes occur when a fault slips abruptly and rapidly. The magnitude of the displacements scales with both the quantity of slip ( $s$ ) and the area ( $A$ ) that slipped, causing persistent deformation



in the surrounding medium (Earth). The seismic potency is the product of these two variables, or more precisely the surface integral  $\iint s dA$  over the fault surface, which is multiplied by the elastic shear modulus of the surrounding material to provide the seismic moment  $M_0$ . Surface displacements are proportional to the seismic moment for a point source (or a small enough area  $A$ ). Most major earthquakes include slip over a vast fault area, and displacements from finite fault sources are determined by the fault's spatial distribution of slip [5]. Several empirical scaling rules have been observed for the earthquakes. In earthquakes, the seismic moment and amplitude of displacements usually scale with the cube of rupture length-width. This is because the average slip in an earthquake increases with fault size. Large earthquakes, therefore, create substantially bigger displacements than minor earthquakes. Fast seismic rupture, in particular, occurs (for the most part) at shallow depths, implying that rupture regions for major earthquakes are typically much longer than they are wide. Seismology is primarily concerned with the observation and interpretation/modeling of seismic waves that have the potential to spread globally in the event of large earthquakes. Fault slip produces a wide range of seismic waves with different propagation pathways and velocities. Seismic waves can be described in two ways--as body waves, which travel through the Earth, and as surface waves, which travel along the Earth's surface. The two types of body waves are compressional waves (P waves) and shear waves (S waves). Due to their fast travel speed, P waves arrive first. In comparison, surface waves move more slowly, sometimes even across vast distances, and reach even later [3], [9].

### **2.6.1 Real-Time Applications of Earthquake Warning**

Earthquakes that are large enough to cause damage or death can cause massive displacements close to the epicenter. Large subduction zone earthquakes, which are capable of causing deadly Tsunamis, can cause displacements as large as several meters in the near field [10]. Real-time GNSS displacements can provide estimations of earthquake magnitude and rupture dimensions that are extremely useful in conjunction with real-time seismology data. Because the seismic measurements employed are limited in frequency band, some commonly used seismic methods to assess earthquake magnitude saturate (Fig. 2.6). As a result, earthquakes that are significantly larger than  $M_w 8$  can be reported at first. The displacements observed by GNSS, on the other hand, are not hampered by this limitation, and the displacements from a  $M_w 9$  earthquake are immediately recognizably larger than those from a  $M_w 8$  earthquake [3].



**Fig. 2.6.** Different EQ Precursors for seismo anomalies detection [11]

## Limitations

There are a number of possible drawbacks to using real-time GNSS for earthquake early warning. The method's inherent limits include the timing and size of the deformation signal, while the precision of the GNSS orbit and clock products, as well as quality control, can be addressed using improved methodologies [4]. The displacements of a GNSS site will be measured when the seismic body waves arrive at the site, not when the earthquake starts. This means that only GNSS sites located close to the fault rupture can help with earthquake early warning, which aims to warn of ground shaking before it occurs [3].

### 2.6.2 Post-seismic Deformation

Transient displacement following the earthquakes is referred to as post-seismic deformation. Substantial earthquakes modify the pattern and pace of strain near the fault by causing large stress shifts. Earth materials can act as elastic, plastic/ductile, viscoelastic, or viscous materials depending on temperature, pressure, and stress. The ability to measure these attributes and investigate the forces that drive tectonic deformation is greatly enhanced by post-seismic deformation [3], [9].

## **2.7 Seismic Precursors**

The process of earthquake preparation was separated into five stages within one period of the seismic cycle. Various physical properties are investigated. The first (and the one that first proposed the dilatation idea) is the change in P-wave velocity. The build-up of dilatation is indicated by a decrease in velocity in the second stage. Ground uplift and tilt, radon emission, electric resistivity, and a number of tiny earthquakes are some of the other precursors in the area of earthquake preparedness/preparation. On the basis of current understanding, we can claim that these are not the only precursors that have been identified. They are most likely examples of a variety of antecedents or precursors, including mechanical deformation, geochemical and hydrological precursors, electromagnetic precursors, and, of course, seismic precursors. Increased moment release and seismic quiescence before an earthquake are well-known seismic precursors [12]. Based on the temporal and spatial distribution of small earthquakes relative to background seismicity, more and more probable precursors linked seismic pattern and seismic parameters are discovered in earthquake prediction practice. If one incident can be identified as a foreshock, it should be a sign of a future significant earthquake epicenter location. The activity of aftershocks and/or a swarm of tiny earthquakes in a tectonic area is linked to a 'seismic window' as one type of precursor. Future earthquakes are predicted by earthquake migration, co-related earthquakes, and recurrent earthquakes, all of which are intimately tied to geodynamic conditions and tectonic settings in a certain time window/range.

A massive concentration of stress in the Earth's crust should not be surprising to provide precursory indications. Some of these precursors have been linked to a variety of strange events that have been observed on the ground and in the atmosphere. Variations in radon, the electromagnetic field, thermal infrared radiation, outgoing longwave radiation, and the total electron content of the ionosphere have all been used to measure these [12]-[13].

### **2.7.1 Radon Emission**

The relationship between the definition of Radon gas emission [14] and the activation of positive ions[15] can be explained by EQs as a mechanical process that creates abnormalities. During the EQ preparation period, Pulnits [14], claims that various gases, including Radon, were emitted, followed by ion-molecular reactions, cluster formation, and an increase/decrease in the vertical electric field. Through this mechanism, the vertical electric field propagated to lower levels of the ionosphere, and eventually to high altitudes. Due to this propagation, variation in Coriolis effect

and electron density occurs, resulting in the variability of electron concentrations in the F2 layer [16].

### **2.7.2 Positive Hole Emission**

The phenomenon of ionization corresponds with the lithosphere-ionosphere synergy that ionizes the atmosphere near the lithosphere due to the emanation of positive holes from rocks under stress within seismic regions. Degraded solid-state plasma forms at the p-holes, which propagates as a rapidly growing charge cloud when critical charge carrier concentrations are reached. At the air-rock interface, the induction of dielectric breakdown results in corona discharges, light/high-frequency EM radiations upon the charge reaching the surface. [15].

## **3 BACKGROUND AND LITERATURE SURVEY**

### **3.1 How Earthquakes Occurs**

Because earthquakes are caused by the collision of tectonic plates, their worldwide distribution is not uniform and is concentrated at the tectonic plate boundaries. Anderson [9] was the first to introduce the concept of a fault, and he was the first to try to explain the physics of faulting (Anderson) [17] using brittle fracture and the Coulomb criterion, i.e., a linear relationship between normal and shear stresses (or maximum and minimum principal stresses) at failure. The earthquakes happen on a fault, which is a pre-existing discontinuity in the Earth's crust caused by prior crustal motions. In terms of mechanics, an earthquake occurs when the applied stress surpasses the dynamic friction. The rupture criterion was introduced first by Griffith [18]. Following the earthquake, dynamic friction exceeds the stress on the rupture zone.

#### **3.1.1 Earthquake preparation Zone**

The size of EQ preparation zone can be estimated by  $\rho = 10^{0.43M}$  km which is proposed by I.P. Dobrovolsky in 1979 [19]. Earthquake preparation is accompanied by a set of precursory phenomena connected with the transformations within the Earth's crust. In the earthquake preparation zone, gas emanations from the Earth's crust are one of the precursors to earthquakes. As the Earth's crust thin, gaseous emanation from the lower mantle can become a major earthquake precursor. Among the several species originating from the crust, we will examine radon, a highly radioactive gas that can cause the ionization of air. Regular gas discharges, such as CO<sup>2</sup>, can transport geomorphology products including those with high metal concentrations. It is worth



noting that the size of the geochemical precursors aligns with the size of the earthquake preparedness zone. Several days or hours before the earthquake shock, an abnormally high vertical electric field was detected in the earthquake preparation zone, which is usually linked to earthquake preparation [16][20].

### **3.1.2 Reaction of Ionospheric layer to the Earthquake**

#### **Ionosphere**

Solar radiation from both ultra-violet (EUV) and x-ray wavelengths ionizes molecules and atoms in the ionosphere, which then generates an electron layer. It is found in the upper atmosphere between 80 and 600 kilometers (Fig. 3.1). Due to its reflective qualities and ability to change radio signals, the ionosphere is essential for communication and navigation.

#### **D-Layer**

The first experimental result correlating the D-layer reaction with earthquake preparation involved the changes in the properties of VLF signals received by ground-based receivers from remote transmitters, particularly the Omega navigational transmitters. The time of the earthquake was linked to the emergence of VLF signal anomalies.

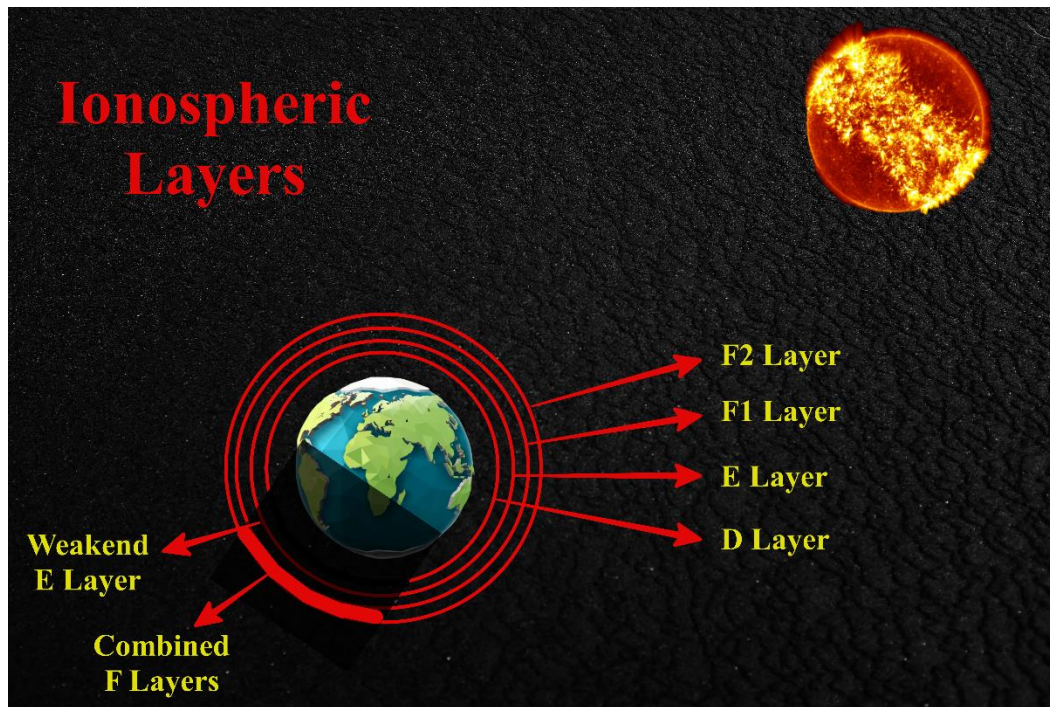
#### **E-Layer**

The essential feature marked in the E-layer is the appearance of considerable scattered or sporadic E-layers across the impending earthquake area many days before the earthquake. Terminator time variations in the occasional E-layer critical frequency produce a rise in electron concentration in the D-layer, which causes the effect of terminator time changes in VLF propagation signals.

#### **F layer**

Strong earthquakes can occupy a considerable section of the ionosphere above the imminent earthquake epicenter projection onto the ionosphere, causing unexpected deviations from regular daily changes in electron concentration [3].

In general, we are interested in the local change of near-earth plasma caused by an abnormal electric field penetrating plasma from the ground level [21].

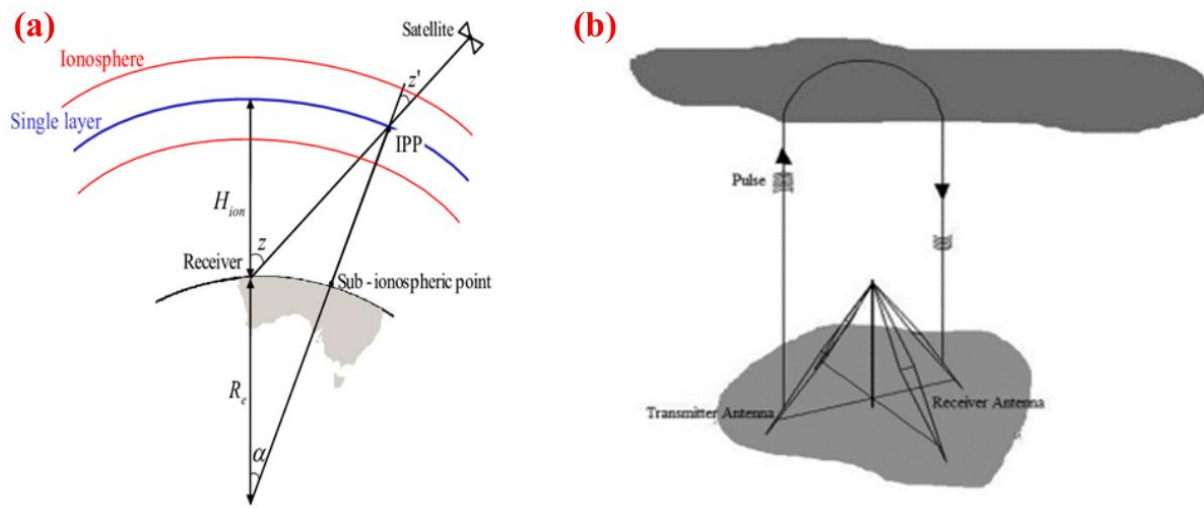


**Fig. 3.1.** Ionosphere structure during daytime and nighttime

### 3.1.3 Ionospheric Anomaly

The Total Electron Content (TEC) can be used to determine the ionospheric disturbance. Essentially, it is a result of the delay caused by the ionosphere when GPS signals propagate, a major cause of error in GPS positioning. Based on the difference between signals on two different frequencies along the ray path between the ground receiver and the satellite, with corrections for the changing length of the ray path as the satellite's elevation angle changes, it is estimated. The Good Friday Alaska earthquake which was occurred on March 27, 1964, is widely credited with igniting seismo-ionospheric coupling research. Pulinets have proposed physical theories for the link of seismicity and ionospheric disturbance[22]. Some of the physical explanations for ground electromagnetic emission have included stress changes in rocks due to piezoelectric and piezomagnetic effects, induction effects due to motion of electric charges in geomagnetic fields, and positive p-hole defects in minerals. Two models have been proposed to link electromagnetic effects at ground level with ionosphere signals: direct connection through electric currents to the lower ionosphere and activation of atmospheric gravity waves by gas releases or temperature anomalies. As a result of adiabatic fluctuations, positive ions combine near the atmosphere and migrate upward to the ionosphere. These complex reactions culminated in the deviation of electromagnetic signals travelling through the ionosphere, which can be detected as ionospheric

anomalies [23]. Statistics have shown that EQs and ionospheric abnormalities are linked significantly at the Chi-Chi earthquake in Taiwan in 1999, for instance [24]. Shah and Jin [25] used GNSS TEC to detect anomalies in ionospheric signals related to distinct responses at different faults highlighted the importance of monitoring SIAs (Seismic Ionospheric Anomalies) associated with  $M_w > 6.0$ . Many researchers have conducted similar research for different regions [26]–[29]. Thus, ionosphere is one of the crucial layers to investigate when attempting to predict earthquakes. Many studies have suggested that SID are caused by an energetic discharge from earthquake preparation zones into the ionosphere. A large magnitude earthquake is predicted based on variations in TEC, foF2, and NmF2, which indicate substantial amounts of energy released during the earthquake preparation period [30][31]. According to statistical analysis of SIP anomalies in the GIM TEC before  $M_w 6.0$  worldwide earthquakes between 2002 and 2010, the frequency of increases and drops in the SIPs increases with earthquake magnitude, but falls with depth and date [32]. Many researches have been reported on the phenomenon of latitudinal TEC augmentation which suggests that an increased amount of energy is transmitted into the atmosphere before a larger earthquake's main shock compared to a smaller one. A small magnitude earthquake, has less evident seismo-ionospheric abnormalities in TEC associated with earthquake breeding zones than a big magnitude earthquake. Ground and satellite measurements of Total Electron Content (TEC), FoF2, and other ionospheric and atmospheric parameters were used to examine the pre and post seismic anomalies generated by large magnitude earthquakes [23][33]–[35] (Fig. 3.2).



**Fig. 3.2.** Ionospheric measurements where (a) Single layer model using GNSS receiver for TEC and (b) using Ionosonde measurement for Fof2

### 3.1.4 Response of Atmosphere to the Earthquake

In the chain of air turbulences, abnormal thunderstorm and atmospheric anomalies prior to the main shock, gas exhalation from the ground and anomalous cloud coverage may be common precursors[36]. There are also reports on elevated ozone concentrations remotely detected before strong earthquakes in Central Asia [37] and in the Nepal region [38]. To account for the ozone formation and the concomitant perturbations in the atmosphere and ionosphere, an emanation of trace gases can be proficient particularly radon, released from the ground due to the build-up of stress. Radon would ionize the air, triggering ozone to form and moisture to condense on the airborne ions, leading to a thermal updraft due to the attendant release of latent heat[39],[40],[20]. Based on laboratory experiments that demonstrated massive air ionization at the rock-to-air interface when rock samples are stressed at one end, Freund et al. [15] anticipated a primarily different process to generate positive and negative airborne ions at the Earth ground-to-air interface prior to major seismic activity. This course starts with the stress activation of peroxy defects in the rocks in and around the hypocenter deep below known as source volume. Peroxy defects exist in igneous minerals in crystalline basement rocks[41],[42]. Their activation leads to the generation of highly mobile electronic charge carriers, called positive holes, which have the remarkable ability to flow out of the source volume, spreading into and through less stressed and unstressed rocks, traveling fast and far, meters in laboratory experiments and kilometers to tens of kilometers in the field [43],[44]. When positive hole charge carriers arrive at the Earth surface, they set up microscopic but steep electric (E) fields [45]. When the E fields reach values in the range of  $10^6 \text{ V m}^{-1}$ , they begin field-ionizing air molecules, in particular  $\text{O}_2$ , which has the lowest ionization potential of the common atmospheric gases, 12.07 eV. This leads to the formation of airborne  $\text{O}^{+2}$  ions. Upon further influx of positive holes to the surface, the E fields reach values adequate to accelerate adventitious free electrons, always available in the air due to cosmic rays, radon, and natural radioactivity from the ground, to energies high enough to impact-ionize neutral gas molecules. This progression then triggers corona discharges, which produce free electrons that attach themselves to neutral gas molecules leading to a massive production of airborne negative air ions. In addition, corona discharges are highly efficient to produce ozone,  $\text{O}_3$ . Freund (2013)[44] further proposed that, when positive airborne ions, such as  $\text{O}^{+2}$ , are generated at ground level, they will rise upward not only because of budding moisture condensation and its release of latent heat, causing a thermal updraft, but also because of a different process that had not yet been

widely considered: due to the polarity of the Global Electric Field positive airborne ions should rise upward towards the ionosphere. Conventionally, the polarity of the Global Electric Field is assumed to be given by the surface of the Earth being negative and the lower edge of the ionosphere being positive [46],[47]. The rationale behind this polarity is that, as lightnings strike the Earth's surface, they would deposit negative charge into the ground negatively and that positive air ions drifting upward above large thundercloud systems would cause the lower edge of the ionosphere to become positively charged [48]. By contrast, the recognition that the Earth surface appears to be always positive due to the presence of peroxy defects in crustal rocks[44] and to the activation of positive hole charge carriers by tectonic stresses [43],[44] calls into question this polarity of the Global Electric Field. If the Earth surface is positive and the lower edge of the ionosphere negative, the estimated 250,000–500,000 V potential difference to the Earth's surface will provide for an effective transport mechanism for positive airborne ions, drifting upward in the vertical E field. This process links positive air ionization at ground level to the perturbations noted in the electron concentration in the ionospheric plasma widely known as TEC anomalies [49].

Conditions created by the suspected massive air ionization are expected to affect the entire epicentral area, which exhibits atmospheric thermal anomalies, vertical winds, aerosol concentration, and other meteorological phenomena[50]. Freund [44] suggested that stress-induced air ionization and formation of positive airborne ions at the ground-to-air interface before major earthquakes, in particular  $O^{+2}$ , causes not only an increase of the overall electrical conductivity in the troposphere but leads also to an upward drift of positive airborne ions towards the stratosphere and mesosphere. This in turn will cause a downward motion of the electrons in the ionospheric plasma and, hence, a TEC anomaly.

Hence summarizing we can say that the positive holes accumulate at the Earth surface, they build up surface charges and associated electric fields, microscopic but strong enough to: (1) massively field-ionize gas molecules, mostly  $O_2$  to produce massive amounts of  $O^{+2}$  ions, which rise through the atmospheric column causing a variety of secondary processes on the way and (2) trigger corona discharges, which produce an abundance of free electrons, negative airborne ions, and ozone.

### **3.1.5 Thermal Anomaly**

In recent years, a growing body of research has been published on the possible atmospheric and surface antecedents of several large-magnitude earthquakes[26],[51],[52]. In recent years, satellite

technologies in remote sensing have been used in pre- and post-seismic monitoring, particularly in measuring thermal anomalies before the earthquake [53]. The scientific community was fascinated by thermal abnormalities in the infrared spectrum from the Moderate Resolution Imaging Spectroradiometer (MODIS), which led to the discovery of EQ precursors in the lower atmosphere near the seismogenic zone during the EQ preparation period. In Russia, China, and Japan, the theory that thermal anomalies are linked to seismic activity has been tested. Using satellite images, Russian researchers noticed thermal anomalies in Central Asia before to an earthquake in 1980[54]. A thermal anomaly is an anomalous spike in Land Surface Temperature (LST) that occurs a few days before an earthquake, boosting temperatures by 3 to 12°C or more, and then diminishes a few days later[55]. The thermal anomalies could be caused by the activation of positive holes (p-holes) in squeezed rock in seismogenic zones during tectonic stress [15]. According to Freund et al. [15] during the preparation stage, an increasing number of p-holes from EQ zones promoted ion activation in the local atmosphere prior to the major shock. EQ energy created considerable heat anomalies above the projected EQ epicenter and accompanying fault lineaments as the major shock day approached. As a result, MODIS-LST imageries provide detailed information about the presence of thermal anomalies linked to EQ. Evident connections of thermal anomalies in LST that appear to be related to pre-seismic activities have been reported in various investigations [56]–[60].

Blackett et al. [61] were arguably the first to suggest that assertions of thermal precursors to earthquakes should be handled carefully. Infrared sensors in modern very high resolution radiometers on remote sensing satellites are used to detect thermal anomalies. The incoming solar and outgoing terrestrial radiations, the characteristics of the surface, and air advection are the key factors affecting fluctuations in ground and air temperature. Heat is absorbed and radiated by the near surface of the Earth. The radiant energy is emitted both during the day and at night, with a peak wavelength of 9.7μm. The rise in temperature is proportional to the specific heat of rocks, dry or wet soil, and the rise in temperature. Air in touch with the ground surface receives the Earth's temperature with a temporal lag. With a temporal lag, air in contact with the ground surface receives the Earth's temperature[62]. As a result, meteorological equipment placed in the Stevenson screen above the Earth's surface are used to study temperature anomalies connected with the surface. Satellite thermal infrared multispectral scanners typically record images in six bands ranging from 8.6 to 12.2μ m, which are used to calculate OLR values (W/m<sup>2</sup>). Shah et al.

[63] compared MODIS satellite derived maximum and minimum temperature data with surface observations given by the India Meteorological Department , finding RMSE of roughly 2.2°C and a mean absolute error of less than  $\pm 2^\circ\text{C}$ .

White–Newsome et al. [64] compared satellite-derived land surface temperature to in situ stations and revealed alike results of higher relative temperatures within a city, dubbed micro urban heat islands. Saraf and Chaudhury [65] mentioned less errors of  $\pm 1^\circ\text{C}$  during Bhuj earthquake 2001. Similar results might be attained using satellites or surface observations, according to these research. Satellite-based thermal data is also utilized to discriminate rocks, surface moisture, and surface temperature of volcanoes, thermal plumes, forest fires, geothermal zones, sea ice, and earthquake-causing faults. Many theories were proposed by Ouzounov and Freund [57] to explain thermal anomalies prior to earthquakes, including: (i) piezoelectric and elastic strain dilatation theory, (ii) rising fluids leading to seepage emitting warm gases, (iii) transient highs in thermal conductivity profile of subsurface rocks, and (iv) rising water levels (v) CO<sub>2</sub> spreading laterally causing local greenhouse effect, (vi) seismo-ionospheric coupling, and (vii) p-hole activation. The source of this phenomena is unknown; according to Ouzounov et al.[66], one possible explanation is the existence of thermal outgoing radiation as a result of near-ground ionization and latent heat change as a result of changes in air humidity and temperature.

This route has also been postulated to induce thermal anomaly as part of a relationship between tectonic stresses, thermodynamic and electrochemical processes in the atmosphere, and increased mid-IR flux, all of which are part of a family of electromagnetic (EM) phenomena connected to earthquake. Temperature rises as a result of pressure. The fact that when we set a coin on a slab of ice, the ice melts due to the weight of the coin employing pressure on the ice is a good example of this. Stresses building up in rocks in a tectonically active location may lead to an upsurge in pressure and, subsequently, an escalation in temperature. Subsurface pressure can also shrink pore pores in rocks, allowing trapped gases to escape. When greenhouse gases (CO<sub>2</sub>, NO<sub>2</sub>, NH<sub>3</sub>) are released into the atmosphere, they outflow to the lower atmosphere and origins a greenhouse effect on the surface of the earth. Acting tensions may cause rock rupture, which could result in an earthquake[67],[62].

Several of the research have attempted to determine the physical basis for the thermal anomaly that produces LST oscillations. Several notable LST precursory studies that used remote sensing observations and were published as research articles in peer-reviewed journals and conference

proceedings over the last two decades have been discovered and gathered. In the Kamchatka Peninsula, researchers [56] used a combination of air temperature, LST, and well observation measurements to propose a litho-atmo-ionospheric coupling model to describe related seismic processes. This was the first time auxiliary field data was combined with remote sensing observations in a holistic study. The MODIS sensor is carried by the Terra and Aqua satellites and provides nearly continuous daily worldwide coverage in 36 spectral bands ranging from 0.415 to 14.235  $\mu\text{m}$ , with spatial resolutions ranging from 250 m (2 bands), 500 m (5 bands), and 1000 m (29 bands). When analyzing the mid-IR emissions prior to an earthquake, Ouzounov and Freund (2004)[57] were among the first researchers to use the MODIS LST product solely. They looked at an extraordinarily devastating M7.7 earthquake that struck Gujarat (India) on January 26, 2001. They noticed a 3–4 °C thermal anomaly that developed 5 days prior to the occurrence. They used the LST generated from MODIS bands 31 and 32, which have bandwidths of 10.78–12.27 m. Another important part of their research was to give more laboratory proof of p-hole activation as a cause for such an LST abnormality [31],[41],[44],[57].

Choudhury Saraf and Choudhury (2005) [58] used AVHRR data and reported the beginnings of this anomaly 12 days previous to the Gujarat earthquake, with a thermal range of 5–7 degrees Celsius. The magnitudes of the LST anomaly recorded in both investigations were similar, however the precursor's inception time differed. This is due to the fact that these research took very different techniques. In a similar series of studies, Ouzounov et al. (2006)[66] performed a more detailed analysis of 6 major earthquakes ( $M \geq 6.8$ ) between 1999 and 2003. They utilized multisensory data (Landsat, NOAA-AVHRR, Terra MODIS, Aqua MODIS) to observe the pre-earthquake LST anomaly and used geosynchronous weather satellite images (GOES-10) to better interpret the results. Ouzounov et al. conducted a more extensive investigation of six significant earthquakes ( $M \geq 6.8$ ) between 1999 and 2003 in a similar set of research. They observed the pre-earthquake LST anomaly using multisensory data such as NOAA-AVHRR, Landsat, , Terra MODIS, Aqua MODIS and geosynchronous weather satellite images i.e., GOES-10 to infer better results (Table. 3.1).

**Table. 3.1.** A list of significant LST precursory studies using remote sensing and published as complete research articles in peer reviewed journals. The articles mentioned here were published between 2007 and 2021 November 2015.

| Earthquake | Study Area | Magnitude | Precursor | Satellite | Reference |
|------------|------------|-----------|-----------|-----------|-----------|
|------------|------------|-----------|-----------|-----------|-----------|



| Date                                         |                                             | Appearance          |                  | Sensor                                    |                                       |
|----------------------------------------------|---------------------------------------------|---------------------|------------------|-------------------------------------------|---------------------------------------|
| 8 oct 2005                                   | Muzaffarabad (Pakistan)                     | 7.6                 | 7 days before    | Terra<br>MODIS                            | Panda et al. (2007)[68]               |
| June 2002–<br>June 2006                      | Iran                                        | 5.8–6.6             | 1–10 days before | NOAA-<br>AVHRR                            | Saraf et al. (2008)[69]               |
| Nov 2001<br>-May 2008                        | Western<br>China                            | 7.3–9               | ~2 weeks after   | Terra<br>MODIS                            | Ma et al. (2010)[70]                  |
| 26 January 2001                              | Gujarat (India)                             | 7.6                 | 9 days before    | Terra<br>MODIS                            | Blackett et al. (2011)[61]            |
| 26 Dec 2003,<br>22 Feb 2005,& 31<br>Mar 2006 | Bam, Zarand, Borujerd (all in Iran)         | 6.1–6.6             | 1–20 days before | Terra<br>MODIS                            | Saradjian and Akhoondzadeh (2011)[51] |
| 20 May 2012,<br>29 May 2012                  | Italy                                       | 5.8, 5.9            | 3 weeks before   | MMERRA<br>Terra<br>MODIS<br>Aqua<br>MODIS | Qin et al. (2012)[40]                 |
| 16 April 2013                                | Saravan (Iran)                              | 7.7                 | 4 days before    | MODIS                                     | Akhoondzadeh (2014)[71]               |
| Jun 2000<br>Sep 2013<br>Oct 2013             | Kharan<br>Awaran<br>Surab (All in Pakistan) | 6.0,<br>6.8,<br>5.2 | 6-8days before   | MODIS                                     | Shah et al. 2021[72]                  |

### 3.2 LAIC Model

If atmospheric and ionospheric manifestations accompany pre-rupture events, it stands to reason that the mechanical action inside the rock column induces non-mechanical responses that appear at the Earth's surface and above, resulting in atmospheric and ionospheric abnormalities. Electrical reactions are the only non-mechanical reactions that can reach this far and generate secondary processes in the atmosphere and ionosphere, according to physics. On a global and local scale, however, there are numerous research on EQ-induced ionospheric and atmospheric abnormalities. [73],[74].

According to Freund [44], the energy released during the tectonic plate motion carries

electromagnetic emissions over a wide frequency range. The emissions consist of the highly mobile charges named positive holes that travel rapidly towards the surface of the Earth during inter-tectonic stress. As more and more positive holes gather at the surface of the Earth, when their cumulative number density exceeds a threshold value, they initiate the air ionization and ions recombination with a strong local electric field at the lower atmosphere. This air ionization leads to variate the ionospheric electron density that affects the radio wave propagation through the ionosphere. In another analysis, Daneshvar and Freund, [75] further explained the propagation of these positive ions in the atmosphere. They suggested that the rise in the positive airborne ions through the atmosphere is primarily attributed to the thermal up-drift caused by the latent heat energy produced by the moisture condensation and vertical penetration of the global electric field. This upwelling electric field then broke up into cells, at the upper atmosphere, and further proceeds towards the ionosphere, ultimately causing ionospheric plasma variations.

Pulinets et al. [39] proposed an Earth's Lithosphere-Atmosphere-Ionosphere Coupling (LAIC) phenomenon by explaining the synergy of several processes that may occur during a seismic event. Their proposed mechanism is largely based upon the emanation of the radon gas that plays a key role in the initialization of the several chemical and physical processes in the atmosphere and, consequently, disturbs the ionospheric electron densities. They suggested that enhanced radon secretions are produced by stressed effects of rocks inside Earth's crust during tectonic activity. These radon increments initiate several physical and chemical processes at the Earth's surface where other gases such as methane, CO<sub>2</sub>, Hydrogen, and Helium serve as carriers. These radon increments produce ion nucleation by ionizing the lower atmospheric air. The production of air ionization is a result of the condensation of the vapor vapors that further releases the latent heat of vaporization to the upper atmosphere. Meanwhile, a significant reduction in the mid-tropospheric relative humidity occurs as a result of this air ionization. This reduction in the relative humidity amplifies the condensation process and enhances the production of the latent heat, which further produces a heat spot at the top of the atmosphere called outgoing longwave radiations. The formation of the large ion clusters due to these atmospheric circulations changes the atmospheric boundary layer conductivity. The currents produced by this effect further varies the ionospheric plasma concentrations. Furthermore, the production of these charge carriers at the atmospheric boundary layer is accompanied by the ascension of charge carriers over the surface of the Earth, which produces the vertical electric field. This electric field uplifts vertical current flow in the

atmosphere, which further leads to enforcing upward-downward flows in negative and positive ions, respectively, in the ionosphere, over the EQ preparation zone. These current dynamos can drive plasma bubbles which might lead to TEC variations and may cause 1- 20% changes in detrended-TEC (dTEC). Explaining the atmospheric currents in the LAIC phenomenon, Sorokin and Ruzhin [76] stated that: during the geological processes (such as an EQ), the intensive processes initiated at the lithosphere and lower atmosphere have an electro-dynamical impact on the ionospheric electron cloud. They further explained the involvement of the quasi-static electric field between Earth's surface and the ionosphere, during the seismic activity, as a vertical structure amplifying the electrical conduction along the altitudes through the ionospheric layers. Their calculations showed that: the quasi-static electric field may attain a value of  $\sim 10$  mV/m in the ionosphere, whereas its value at the ground level is  $\sim 100$  mV/m.

Various investigations on the coupling of the lithosphere, atmosphere, and ionosphere in pre- and post-seismic impacts for various regional and global EQs have been conducted successfully [39],[77],[78]. Moreover, during the seismic preparation phase, Sorokin and Ruzhin [76] described the upward propagation of EQ energy to the lower atmosphere and then to the ionosphere as an electrodynamic impact over the epicenter. They also revealed how the quasi-static electric field between the Earth's surface and the ionosphere played a role in the EQ preparation phase. Similarly, variations in the electric field at various altitudes in the ionosphere's vertical structure result in an anomalous electron cloud over the seismogenic area.

Shi et al. [79] used a dense GPS network of 275 receivers to explore the lithosphere-ionosphere coupling before the massive EQ of the 2015 Mw 7.8 in Nepal, reporting both positive (on the 2nd day) and negative anomalies (on the 14th day) in GPS TEC before the main shock. Similarly, Pikridas et al. [80] looked studied TEC changes in the days leading up to the Mw 6.7 Aegean Sea EQ and discovered a significant drop in TEC values one day before the main shock. They went on to say that these abnormalities show up over the epicenter's position and help to explain the LAIC phenomenon.

## **4 METHODOLOGY**

## 4.1 Study Area

In the conducted study we analyzed two seismic events of  $M_w > 5.3$  in the northern and southern parts of Pakistan (Fig. 4.1). In this research work, the Awaran earthquake of  $M_w$  7.7 which was occurred at 15 km hypocentral depth on September 24, 2013 is studied. Moreover, the epicenter was located at 113 Km North-West of Bela and 61 Km North-East of Awaran on the USGS website [http://earthquake.usgs.gov/earthquakes/eventpage/usb000jyiv#general\\_summary](http://earthquake.usgs.gov/earthquakes/eventpage/usb000jyiv#general_summary) (**Latitude:**  $26^{\circ} 09' 60.00''N$  and **Longitude:**  $65^{\circ} 29' 59.99'' E$ ). The maximum felt intensity of VIII (severe) on the Modified Mercalli scale, as on record of United State Geological Survey (USGS) (Fig. 4.2b).

About a minute after the main shock took place, panic gripped the region, resulting in 825 lives being lost and significant infrastructure damage.

The earthquake was followed by a series of strong aftershocks with a major of  $M_w$  6.8 on September 28, 2013. The affected areas were Awaran, Tirtaj, Gashkore, Nok Jo, Parwar, Dandar, and Hoshab. The earthquake occurred due to an oblique-strike slip-type motion. The tremor occurred in all Pakistan and its neighboring countries; Iran, Afghanistan, and India. A cumulative moment of  $5.4 \times 10^{20}$  Nm was released by this rupture.

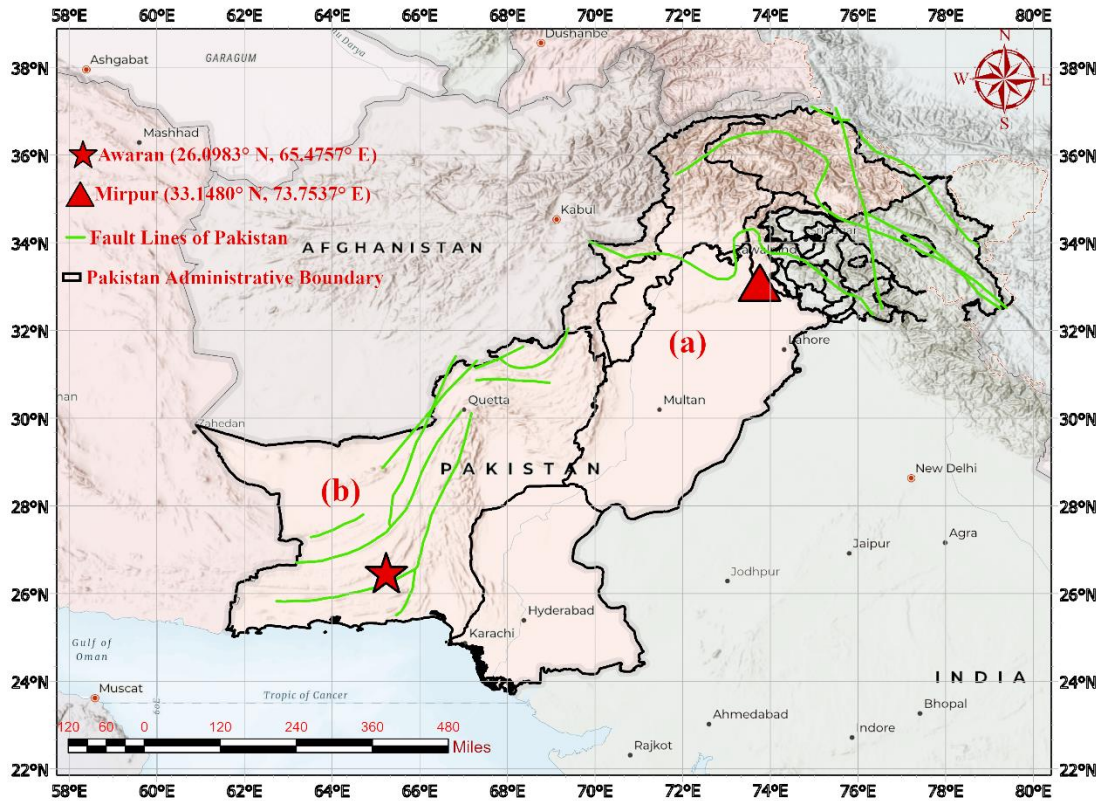
Another case study includes the Kashmir EQ of shallow hypocentral depth 10 km struck the Pakistani region on September 24, 2019. Its epicenter was located at 7 km SSE of Mirpur, Pakistan ( $33.1^{\circ}N$ ,  $73.7^{\circ}E$ ) (Fig. 4.1). The maximum felt intensity of VII (very strong) on the Modified Mercalli scale, as on record of United State Geological Survey (USGS) (Fig. 4.2a). There have been serious damages both to the lives of people and the infrastructure in Mirpur District, resulting in 40 deaths and more than 850 injured.

Further, a number of bridges and parts of many roads have been demolished, especially within 14 km of the main seismic zone. Epicenter of the earthquake was found 3km south of Mirpur and 20km north of Jhelum. More information about this EQ is available on the USGS website (<https://earthquake.usgs.gov/earthquakes/eventpage/us60005mqp/executive>). To check the detailed information about selected EQs see Table 4.1.

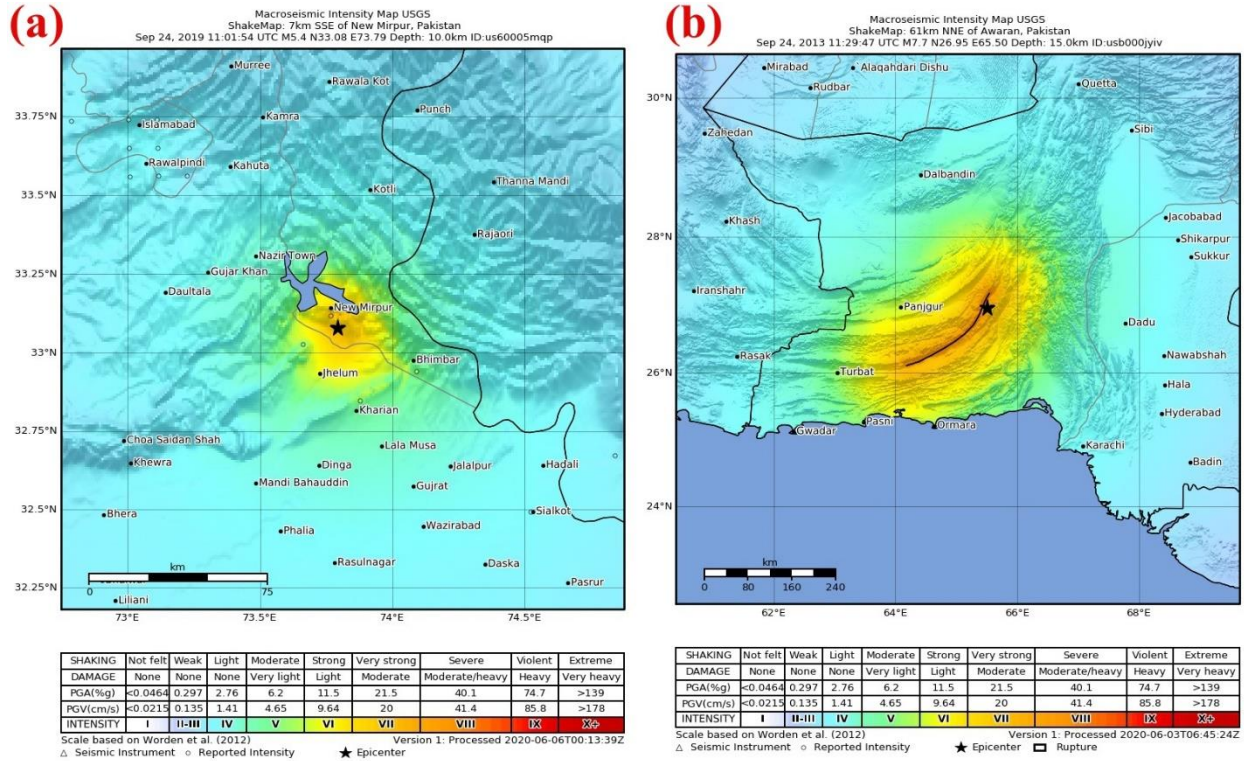
**Table. 4.1.** Geographical locations and other information of the EQ events

| Event | Region | $M_w$ | Date (Time) | Lat (Long) | Depth (km) | Deborowski Region (km) |
|-------|--------|-------|-------------|------------|------------|------------------------|
|-------|--------|-------|-------------|------------|------------|------------------------|

|   |        |     |                                    |                            |    |        |
|---|--------|-----|------------------------------------|----------------------------|----|--------|
| 1 | Mirpur | 5.4 | 24 Sep, 2019<br>(11:01:54.990 UTC) | 33.1480° N<br>(73.7537° E) | 10 | 209.8  |
| 2 | Awaran | 7.7 | 24 Sep, 2013<br>(11:29:47.970 UTC) | 26.0983° N<br>(65.4757° E) | 15 | 2046.4 |



**Fig. 4.1.** Study area map where red star shows Awaran EQs (24 Sep, 2013) epicenter and red triangle is the location of Mirpur EQ (24 Sep, 2013) epicenter while green lines represents fault lines of Pakistan

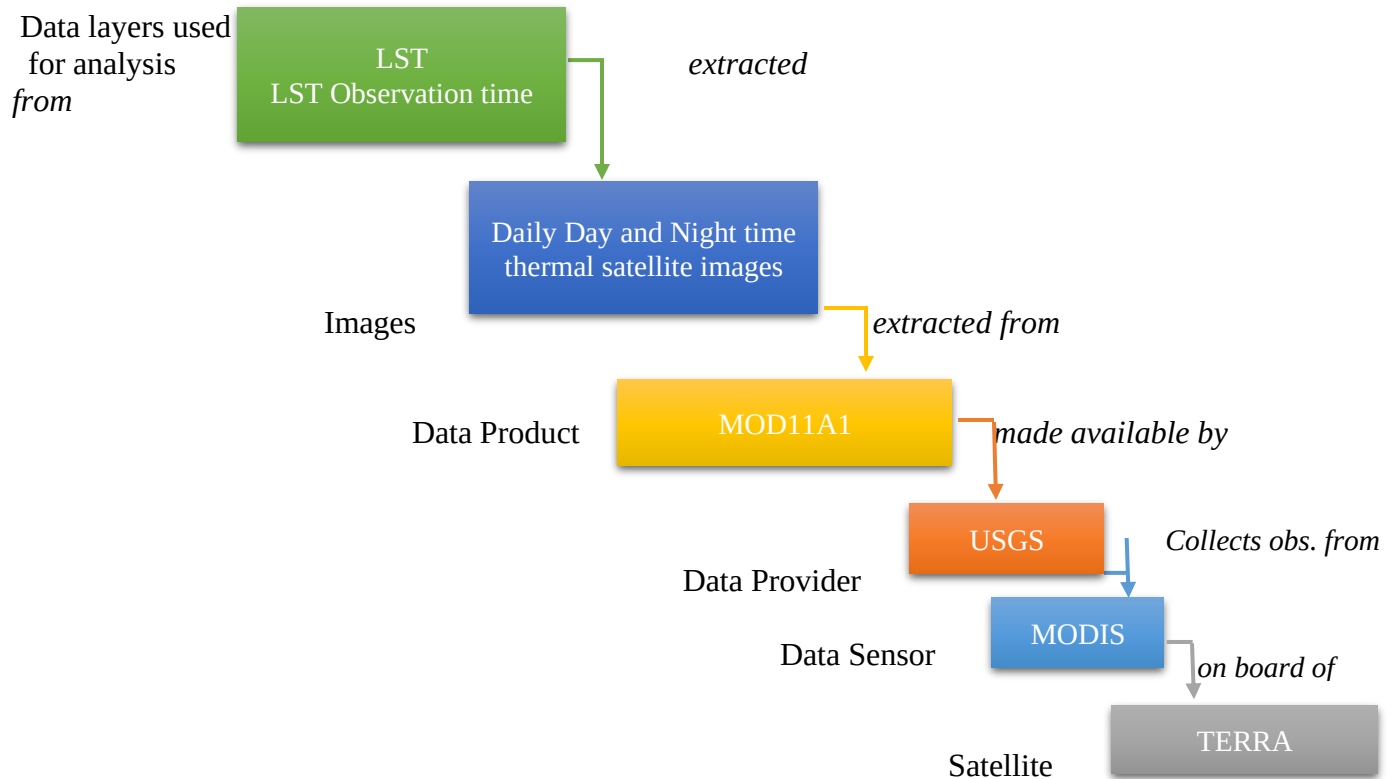


**Fig. 4.2.** USGS Shake map (a) Mirpur (24 Sep, 2019) (b) Awaran (24 Sep, 2013)

## 4.2 LST Retrieval

Daytime and nighttime LSTs are computed for both earthquakes for the time period of 2 months before and 2 months after the main shock day. The land surface temperature data were derived from daily satellite images taken during the day and at night (Fig. 4.3). The USGS makes the satellite images available through MOD11A. LST is retrieved using a generalized split window algorithm. Based on the differential absorption in adjacent infrared bands, the split window algorithm corrects for atmospheric effects. Specifically, in the case of LST, these adjacent bands are 31 and 32. The MODIS sensors collect thermal information through these two bands. Depending on how they are configured, the sensor channels have bandwidths between 10.780 - 11.280 m and 11.770 - 12.270 m. Channels like these are referred to as atmospheric windows since they can receive wavelength-specific radiation from earth with minimal signal loss. A grid format of these signals is then created using digital numbers, which are then distributed as images. Images contain not only land surface temperature data, but also accuracy indicators, time and angle of observation on a pixel basis and other relevant information. Information about the images and

other metadata, such as text, are part of this data. For this research, the MOD11A1 product image layers used are (a) LST\_Day\_1km (b) LST\_Night\_1km and (c) Night/Day -view-time. Image dimensions are 927 by 927 meters with a spatial resolution of 1200x1200. Depending on emissivity, atmosphere content, sensor view angle, etc., LST accuracy varies by an average of 1 K.



**Fig. 4.3.** Step by step retrieval of LST satellite image acquisition

#### 4.2.1 Pre-processing of Satellite Images

Pre-processing included projecting all downloaded images to Universal Transversal Mercator (UTM) projection, Zone 48, North, and clipping to the study area. The re-projection Tool in ArcGIS was used for this task. MODIS user product guide specifies the scale coefficients to be used for converting image values to Kelvin. Our research interests include possible anomalous LST patterns within a given time period and within a specific space relative to the epicenter of the earthquake. The epicentral distance, along with the day on which the LST observations were acquired, are two pieces of information that are crucial.



EQ variations over the epicenter detected in the temporal analysis are scrutinized on the spatial images. The effective area observations are considered to prepare time series maps of spatial mean LST for day and nighttime.

### 4.3 TEC Estimation

For analyzing Ionospheric parameter e.g. TEC during Awaran EQ 2013 we received data from IONOLAB while GPS station was installed on Sukkur (N 27° 42' 18.576", E 68° 51' 26.568") in 2019 so we retrieved TEC data of Mirpur EQ 2019 from the Sukkur GPS station and examined the ionospheric fluctuations around the EQ epicenter (Fig. 4.4). In the form of VTEC, the TEC before and after the main shock is examined along the line of sight from satellite to receiving station. Slant TEC (STEC) data are available in Ionospheric Scintillation Monitoring Records (ISMR) and are acquired using a multi-frequency GNSS receiver, Septentrio Polarx5S. STEC can be obtained along the view line between the transmitter and the ground station. A TECU is defined as the number of free electrons in a (1 x 1) square meter section (where one TECU equals  $10^{16}$  electrons/m<sup>2</sup>). IONOLAB software is used to calculate STEC. Further, the VTEC is processed from STEC as below.

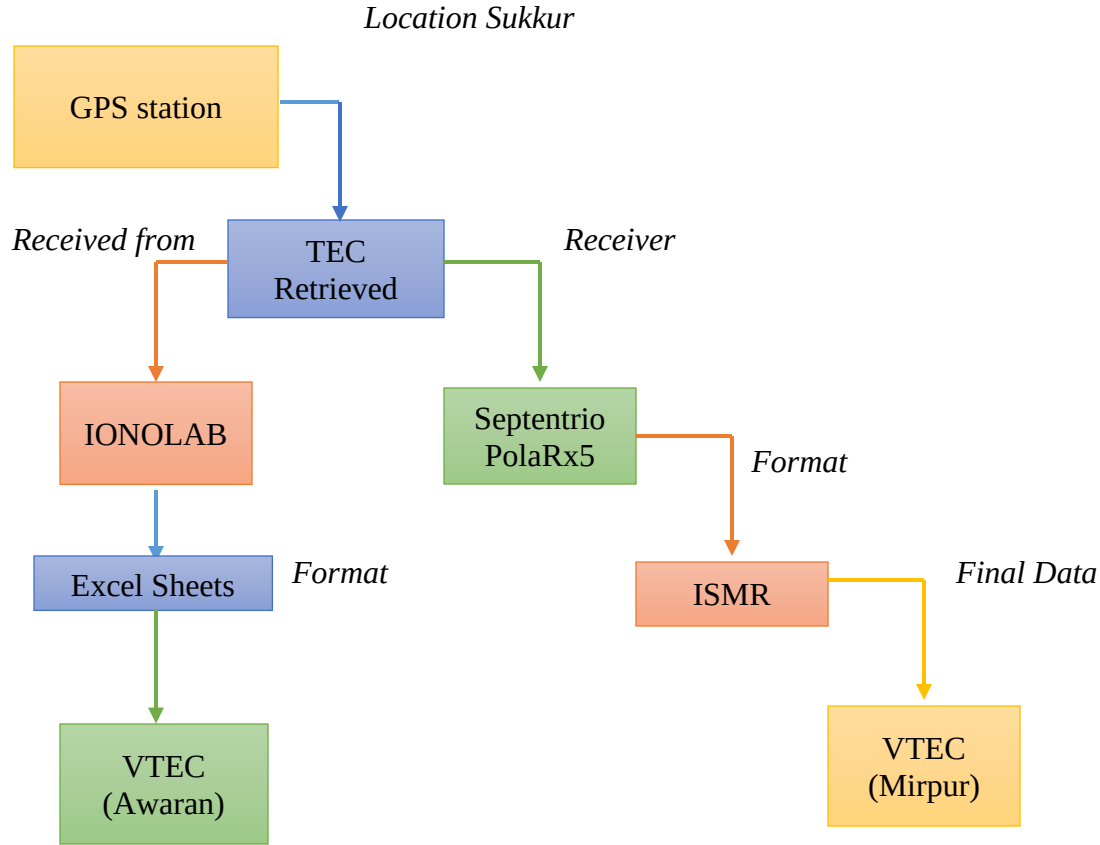
$$\text{STEC}_a^h = - \frac{f_1^2 f_2^2}{40.3(f_1^2 - f_2^2)} (P_{4,a}^h - c \times \text{DCB}_a - c \times \text{DCB}^h) \quad (1)$$

The equation (1) corresponds dual frequencies from GPS as  $f_1$  and  $f_2$  and the speed of light is depicted as  $c$ . Furthermore,  $P_{4,a}^h$  shows differences between smoothed coded measurements, while  $\text{DCB}_a$  refers to satellite and  $\text{DCB}^h$  refers to receiver are differential code biases. To get VTEC from STEC, we need to use the below equation on STEC.

$$\text{VTEC} = \text{STEC} \times \cos\left(\arcsin\left(\frac{R \sin z}{R+H}\right)\right) \quad (2)$$

Where  $H$  is ionosphere top layer height and it is 350 km in the case of Klobuchar (1987) model for VTEC estimation. Moreover,  $R$  and  $z$  are the Earth's radius and zenith angle of the satellite, respectively. VTEC is denoted in TEC unit (where, 1 TECU =  $10^{16}$  el/m<sup>2</sup>). The GPS TEC and MODIS LST data are designed in the critical zone of the seismogenic region.





**Fig. 4.4.** Diagram of VTEC acquisition

By estimating the thermal reactions of seismic activity, an earthquake preparation zone can be identified.

The earthquake preparation zone can be estimated by the below relation.

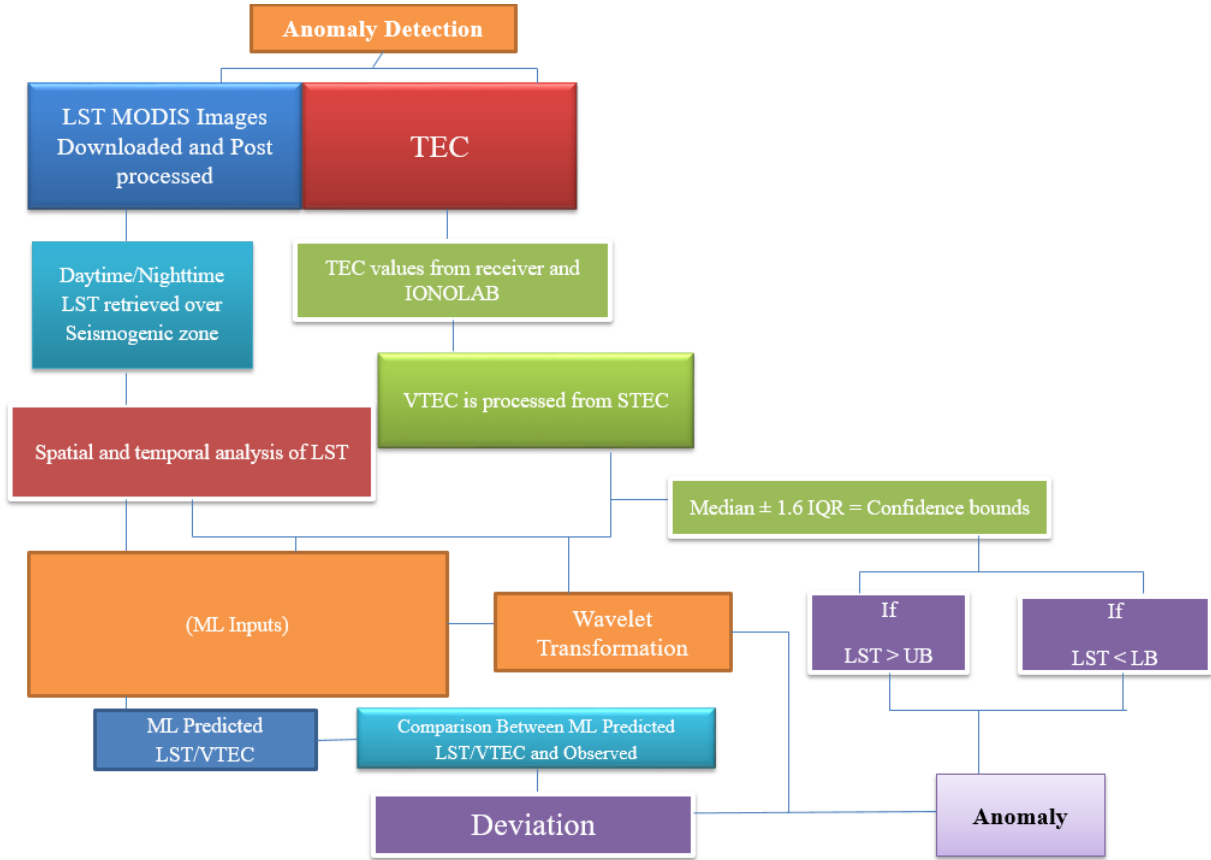
$$RD = 10^{0.43 \times M} \quad (3)$$

where, RD is the radius of an earthquake preparation zone and M is the magnitude of the seismic activity[19]. The temperature anomalies from MODIS LST are estimated above the epicenter in Dobrovolsky's region in this research. When detecting TIR anomalies in the earthquake preparation zone, the anomalies will be more obvious closer to the epicenter than further away[56],[59],[66]. Analyses of TIR data generally use Latitude and Longitude. The size of the TIR data can also vary. In this study, the varying duration of LST within the main shock preparation zone is studied by using the satellite Infra-Red imagery recorded by the Moderate

Resolution Imaging Spectroradiometer (MODIS) on the Terra satellite (December 18, 1999). The National Aeronautics and Space Administration (NASA) provides data, (<http://modis.gsfc.nasa.gov/data>) daily in 36 high-spectral bands covering 2300 km from the earth's surface [66]. For the current analysis of TIR data, daytime and nighttime LST are extracted for Awaran 2013 and Mirpur 2019 earthquakes within a 2 month period prior to and 2 months subsequent to its main occurrence date. As a way of demonstrating the validity of the temporal LST anomaly, the spatial maps of two months before and two months after an earthquake are mentioned. For both daytime and nighttime, Fig 3 and 4 display sample plots to demonstrate fluctuations in LST prior to and after the EQ event occurring within 10 days before and after the EQ.

#### **4.4 Implemented Methods**

A number of conventional and machine learning techniques exist for detecting irregularities in nonlinear time series of TEC and LST data [81]. As part of this study, we investigate the abnormal LST and TEC values using the IQR bounds along with other approaches such as wavelet transformation and machine learning algorithms as standard, classical, and intelligent methods. If the processed values exceed predetermined threshold (confidence bounds), anomalous variations in data will be identified as anomaly, [Figure 4.5](#) summarizes the overall proposed methodology to detect anomaly.



**Fig. 4.5.** Flow chart for detecting anomaly in LST and VTEC data using standard, classical and machine learning approaches

#### 4.4.1 Inter Quartile Range (IQR) Method

Interquartile Range (IQR) is one of the most common methods used in statistics. The IQR concept quantifies statistical dispersion and variability in data by dividing the dataset into quartiles. The four data intervals are determined according to the values of the data and their relation to overall datasets. Four intervals and three points are drawn from the data by a quartile.

A Gaussian probability function is usually not implemented for non-linear values, so few researchers use the IQR and median methods i.e., Liu et al.[49], Akhoondzadeh [82] etc. The IQR of data is used to specify the bounds i.e., higher and lower in order to distinguish anomalies from natural variations. The bounds can be calculated using the following equations

$$x_{high} = M + k \times IQR \quad (4)$$

$$x_{low} = M - k \times IQR \quad (5)$$

$$x_{low} < x < x_{high} \rightarrow -k < \frac{x-M}{IQR} < k \quad (6)$$

$$Dx = \frac{x-M}{IQR} \quad (7)$$

where  $x$ ,  $x_{high}$ ,  $x_{low}$ ,  $M$ ,  $IQR$ , and  $Dx$  are parameters, higher bound, lower bound, median value, interquartile range, and differential of  $x$ , respectively. Furthermore, if  $Dx$  exceeds  $k$  ( $|Dx| > k$ ), the behavior of the relevant parameter ( $x$ ) is described as anomalous. According to Eq(6),  $p = \pm 100 \times (|Dx| - k) / k$  specifies the percentage of the parameter change from the undisturbed state. Whenever an observed LST/TEC diverges from one lower or higher bound, we stated that there is a 66% probability that there has been a variation in the data series [49][82]. The lower and upper bounds are calculated from the median and IQR of all the data of 4 months (i.e., 2 months before/after the main shock) for daily LST and TEC values.

#### 4.4.2 Wavelet Transformation Method

As a powerful technique in studying time series, the wavelet transformation has been used in various fields because it is able to extract multiple components, including trend, seasonally based, and abrupt components. In the frequency domain, signals are decomposed and characterized by distribution states in the time domain using wavelet transforms. Wavelet functions were used to widen the time series into the time-frequency space as a result of the CWT analysis in order to find its localized recurring periodicities. Both time and frequency domains are localized with zero means for the wavelet functions. In the current study to locate variability in MODIS LST and GPS VTEC data as anomaly associated with the Awaran and Mirpur earthquakes, the wavelet transformation has been implemented on time series. The CWT revealed a relationship between the continuous-time signal ( $x(t)$ ) and a wavelet-based function and is defined by the formula given below

$$W_{\Psi}(s, \gamma) = \frac{1}{\sqrt{s}} \int_{-\infty}^{\infty} x(t) \Psi^* \left( \frac{t-\gamma}{s} \right) dt \quad (8)$$

Here  $s$  is the dilation of wavelet i.e., scale and  $\gamma$  depicts wavelet translation and specifies the time localization, complex conjugate of the mother wavelet  $\Psi(t)$  is  $\Psi^*(t)$ . The coefficient  $\frac{1}{\sqrt{s}}$  is energy normalized factor. As the wavelet scale and translation factor change along with time, we produce a scalogram. Calculations were carried out to recognize the amplitude of a specific scale and its variations with time. Based on the scale value and the centered frequency of the wavelets, a frequency value was then calculated. In the CWT analysis, several types of wavelet

functions can be used. Amongst these, the Morlet wavelet function provides rational localization in both spaces (frequency and time), and was therefore implemented in the present study.

Equation of Morlet wavelet is given by

$$\text{Morlet wavelet (morl)} : \Psi(t) = e^{-\frac{t^2}{2}} \cos(5t) \quad (9)$$

Morlet wavelet extricated features with equal variance in frequency and time, which provided a pledge of extracting temporal features and verified that the time-frequency resolution can be implemented to signals of interest. It aided in reducing the sharp edges of time series data hence the ripple effects were ignored. However by using components of the wavelet transform, high-frequency noise and low-frequency seasonal components have been eradicated. LST/VTEC high perturbations were observed by wavelet coefficients surpassing the pre-defined value of the threshold. In this study,  $M + 1.6 \times \text{IQR}$  has been designated as optimum threshold values to detect the unusual wavelet coefficient values for the studied earthquake events. Optimal threshold values (k) were defined on the basis of an iterative process. The wavelet coefficients of values of LST/VTEC greater than the thresholds are nominated as anomaly values [82].

#### **4.4.3 Neural Network Methods**

The learning process of complex functional relationships from data or by recognizing patterns can be modeled by an intelligent system such as a neural network (NN). Even if the series leading fundamental laws are complex or unknown, neural networks can capture the autocorrelation structure. The NNs are composed of processing units, while neurons are associated with networks defined by their weighted connections. Through the training process, the weights of the connections in the network are determined. If the prediction error exceeds a predetermined threshold, anomalous variations in measured values will be identified. There are two models i.e., Nonlinear Autoregressive models with exogenous inputs (NARX) and Multilayer Perceptron (MLP) which are corroborated to analyze the observed time series (LST and VTEC).

##### **4.4.3.1 NARX**

In this analysis, recurrent neural network Nonlinear Autoregressive models with exogenous inputs (NARX) model is used to validate anomalies in LST and VTEC time series data. This class of neural networks (i.e., NARX) is verified to be reasonable for signifying nonlinear systems explicitly time series [83][72]. The NARX model for time series estimation is used where the

dependent output signal  $y(t)$  is regressed using both the previous output signal estimations and the independent input signal observations.

The equation of NARX model is showed as below.

$$y(t) = f(y(t-1), y(t-2), \dots, y(t-n_y), x(t-1), x(t-2), \dots, x(t-n_x)) \quad (10)$$

During the training process, we calculated the network configuration and estimated the minimum predictive error, which included effective parameters like output, input lags, number of hidden layers, and corresponding neurons; the activation functions and training algorithm were obtained through iterations. In both hidden and output layers, the ideal outcomes are specified by linear and sigmoid activation functions. For function approximation problems involving networks with a few hundred weights, the Levenberg–Marquardt technique will provide faster convergence [72]. Hence, for training algorithm the Levenberg–Marquardt optimization is applied. Moreover, the implementation of prediction process,  $N$  observations including  $y_1, y_2, \dots, y_N$  and  $y_{N+1}, y_{N+2}, \dots, y_{N+m}$  are selected as training set and test set, respectively. By determining the number of lagged observations that correspond to input nodes, we discovered the underlying patterns of the time series. This study proposes a network consisting of six nodes in the input layer, 10 nodes in the first and second hidden layers, and one node in the output layer (Fig. 4.6).

Input layer neurons receive information from multiple sources through observed values. For this analysis, corrected LST values, their corresponding time, and LST deviation from mean values are used as input values. For VTEC data along with time information we used its deviation from mean VTEC as input parameters. Each layer of hidden neurons is given its weight by the preceding input layer. An effective combination of hidden neurons activates the transfer function. The proposed network's training patterns are below.

$$y_4 = f(y_1, y_2, y_3, t_1, t_2, t_3) \quad (11)$$

$$y_5 = f(y_2, y_3, y_4, t_2, t_3, t_4) \quad (12)$$

$$y_N = f(y_{N-3}, y_{N-2}, y_{N-1}, t_{N-3}, t_{N-2}, t_{N-1}) \quad (13)$$

The prediction performance assessment is done to minimize the prediction error (PE) by finding optimized weights. PE equation can be written as:

$$PE = \sum_{k=0}^N (\hat{y}(t-k) - y(t-k)) \quad (14)$$

$\hat{y}$  is the network's output.

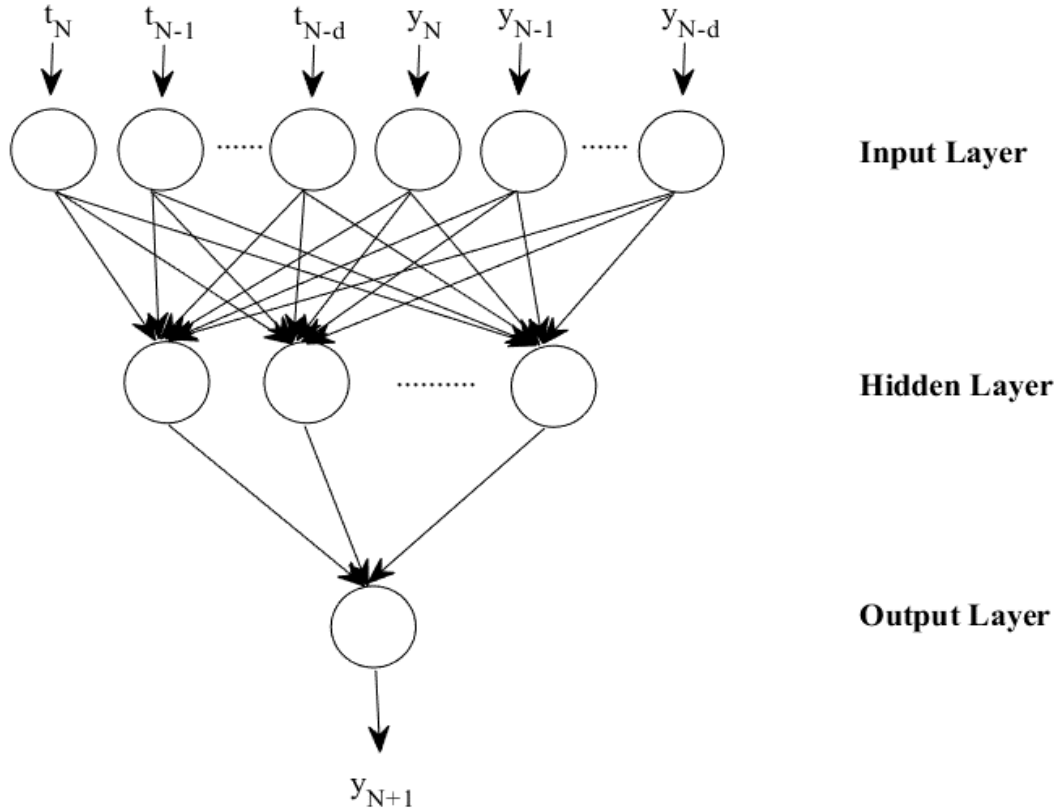
The testing patterns are:

$$y_{N+4} = f(y_{N+1}, y_{N+2}, y_{N+3}, t_{N+1}, t_{N+2}, t_{N+3}) \quad (15)$$

$$y_{N+5} = f(y_{N+2}, y_{N+3}, y_{N+4}, t_{N+2}, t_{N+3}, t_{N+4}) \quad (16)$$

$$y_{N+m} = f(y_{N+m-3}, y_{N+m-2}, y_{N+m-1}, t_{N+m-3}, t_{N+m-2}, t_{N+m-1}) \quad (17)$$

Finally, the anomaly is perceived by assessing the difference between predicted and actual values exceeding the predefined bounds.



**Fig. 4.6.** NARX neural network for time series data prediction

#### 4.4.3.2 Multilayer Perceptron

A feed-forward Multilayer Perceptron (MLP) network, in which neurons are arranged in layers and only forward connections exist, is the most successful and popular model. This model consists of  $n$  input units,  $k$  hidden units, one output unit, connection weights  $w_{ij}$  from input  $i$ th unit and hidden  $j$ th unit, and connection weights  $T_j$  from hidden unit ( $j$ th) to output unit. Network hidden layer consists of nodes, which are attached to both input and output. The total data is divided into two sets: training and testing to detect anomalies.

The build of neural networks and the measurement of their PE require training and test sets. The

process of training determines how weights are assigned to network connections. As long as the prediction error value exceeds the bounds, the model measured values will be considered anomalous. The best activation function for output layers is linear function, according to our analysis.

In all hidden nodes, the Tan sigmoid function purposes as the transfer function. The tan sigmoid function is:

$$f(x) = \frac{2}{1 + e^{-2x}} - 1 \quad (15)$$

Levenberg-Marquardt optimization was selected as the training algorithm. For the implementation of prediction process, N observations including  $y_1, y_2, \dots, y_N$  and  $y_{N+1}, y_{N+2}, \dots, y_{N+m}$  are selected as training set and test set, respectively. This model also uses the time series pattern expressed by the number of lagged observations that correspond to inputs, similar to NARX. A variety of input nodes affect the network's predictive or learning ability.

Different input nodes can affect either the learning or predictive capability of the network [55][84]. In this study, input layer, hidden layer and output layer consist of three, two and one node, respectively (Fig. 4.7). The contemplated inputs includes same parameters of LST and VTEC time series as mentioned in section 4.4.3.1. A pattern vector is established for every four observations in the training set, out of which three correspond to input values and one corresponds to output values.

Proposed network's training pattern are:

$$X_4 = f(x_1, x_2, x_3) \quad (16)$$

$$X_5 = f(x_2, x_3, x_4) \quad (17)$$

$$X_N = f(X_{N-3}, X_{N-2}, X_{N-1}) \quad (18)$$

Execution of training process is done to minimize the PE by finding the optimized connection weights. PE equation can be written as:

$$PE = \sum_{i=4}^N X_i - \hat{X}_i \quad (19)$$

$\hat{X}_i$  is the network's output.

Patterns for testing are:

$$X_{N+4} = f(X_{N+1}, X_{N+2}, X_{N+3}) \quad (20)$$

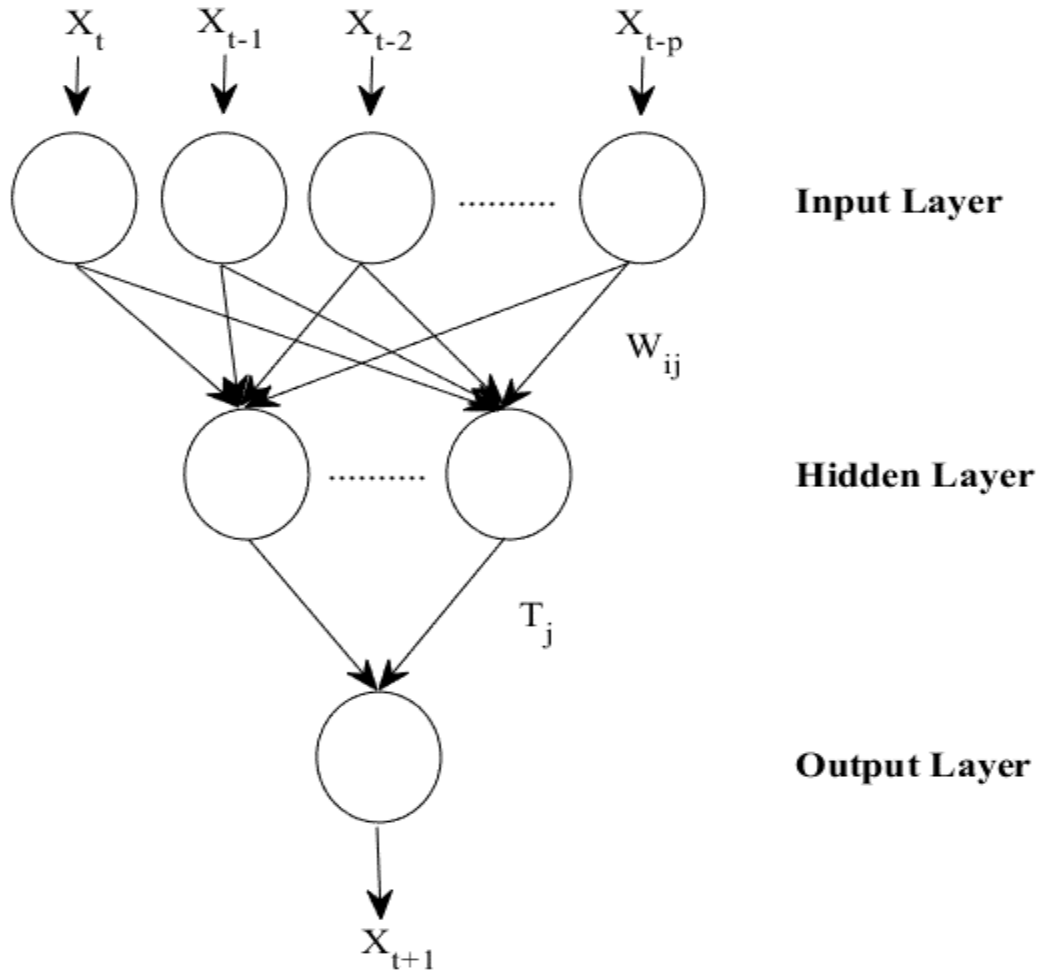
$$X_{N+5} = f(X_{N+2}, X_{N+3}, X_{N+4}) \quad (21)$$

$$X_{N+m} = f(X_{N+m-3}, X_{N+m-2}, X_{N+m-1}) \quad (22)$$

In the case of testing process, difference of values ( $DX_i$ ) between actual ( $X_i$ ) and predicted ( $\hat{X}_i$ )



values exceeding pre-defined bounds are considered as anomaly.



**Fig. 4.7.** MLP feed forward neural network for time series prediction

## 5 RESULTS AND DISCUSSION

### 5.1 Discussion

Pakistan is one of Asia's most earthquake-prone countries, with a high density of active faults. The site is located at the meeting point of three plates: the Indian, Eurasia, and Arabian plate borders. In the southern region, subduction zone exists along Makran coast. The Makran plate border is one of the major deformation zones in southwestern Pakistan and southern Iran. The Makran Subduction Zone is mostly defined by transpressional strike-slip systems. A fault system called Ornach-Nal marks the eastern limit, while Minab-Zendan marks the western one. Located on the

left-lateral transform plate boundary, the Chaman fault system separates the Indian and Eurasian plates. Ornach Nal, Ghazaband, and Chaman sinistral strike slip faults make up the Chaman fault system. The Chaman fault is joined by the Makran faults Siahan, Panjgur, and Hoshab. They are responsible for shearing between India and Eurasia, as well as a shortening caused by Arabian-Eurasian convergence.

Freund [44] suggested that before big earthquakes, stress-induced air ionization and creation of positive airborne ions, particularly  $O^{+2}$ , causes an increase in total electrical conductivity in the troposphere as well as an upward drift of positive airborne ions into the stratosphere and mesosphere. This, in turn, will induce electrons in ionospheric plasma to move downward, hence a TEC anomaly. Moreover thermal anomalies reported a few days before an earthquake [56][85] were commonly understood as thermal flux deposited from the earth's crust in seismically active areas [57]. Many pre-earthquake signals point to processes taking place at the Earth surface or the ground-to-air interface. The reported phenomena include – Various atmospheric effects, – Perturbations in the ionosphere, – Luminous phenomena, often called earthquake lights, – Enhanced infrared emission from around the epicentral region seen in satellite images, – Radon emanation from the ground.

During our studied parameters these physical parameters showed abnormal characteristics in the lithosphere and subsequently, in the atmosphere prior and even after the main events as well as a TEC anomaly in ionosphere.

### **5.1.1 LST Anomaly**

In this study, the LST anomalies are investigated 2 months before and after the two EQs in the southern and northern territory of Pakistan by examining the temporal series of daytime and nighttime MODIS LST images. Where we observed anomalous behavior of LST as EQ precursor and even as EQ response within 10 day window. Taking in to account the post-EQ anomaly or EQ response we observed an escalation, the mechanism is unclear and the debate is required, however we infer this phenomenon to be most likely due to the forthcoming major aftershocks. Most earthquakes produce aftershocks because the rocks on both sides of the earthquake fault have migrated to new positions, transferring stress to neighboring faults. The other faults generate aftershocks when they move to release the new stress. The length of time that aftershocks last depends on the size of the main earthquake.

The physical mechanism encompasses that these aftershocks usually occur in rocks near the epicenter or along the fault line where the primary earthquake occurred. This transmission either creates new stresses or enhances existing tensions within or between rocks. A series of lesser tremors are produced when a sudden accumulation of stress is significant enough to fracture these rocks, alleviating the stress between them. Following an earthquake, small tremors are reported as if they occurred within the fault that caused the earthquake or if they occurred outside the fault within one fault length (the length of the fault from end to end) of the epicenter. The magnitude and frequency of aftershocks decrease over time. In general, the rate of decline inversely scales with the amount of time passing since the principal earthquake. In an earthquake's aftermath, the sequence of aftershocks ceases as soon as the aftershock rate reaches pre-earthquake levels.

The anomalies are verified using IQR, Wavelet Transformation and Neural networks counting NARX and MLP techniques. Our data do not include variations of other geophysical events, which were found to be insignificant. Atmospheric variations caused by other geophysical events are negligible. To provide a stringent proof for the relationship of MODIS LST to the  $M_w > 5.4$  earthquakes, the spatial images were used. We obtained the normal temperature in  $^{\circ}\text{C}$  for the MODIS LST. The sinusoidal map projection of LST images was converted to specific geographic latitude/longitude under the Datum WGS-84 platform. The resolution of each pixel was enhanced to 500 times by performing this step. Summarizing the analysis we also shadowed the point of view hypothesized by Pulinets and Ouzounov [33] where they examined the characteristics of radon emission in seismically active regions, positing that a radon release increases the LST due to vertical electric fields, ultimately causing thermal anomalies.

### **5.1.2 TEC Anomaly**

The seismo ionospheric anomalies related to EQs are scrutinized by considering the data from GPS TEC over the epicenter for both EQs and anomaly is substantiated using IQR, Wavelet Transformation and Neural networks counting NARX and MLP techniques. We assessed the irregularity in TEC within 10 day window before and after the EQ commencement.

### **5.1.3 Origin of Southern and Northern Earthquakes**

On the Hoshab fault, the 2013 Awaran earthquake was a strike-slip event. There was an average slip of 4.2 m along the main trace of the fault and a maximum local offset of 11.4 m. A maximum

displacement of 16m was recorded. Makran accretionary complex contains the Hoshab fault. Arabia and Eurasia reach convergence at a rate of 3 cm/yr in Makran convergence zone.

This earthquake is one of the enumerated slip events to occur on a curved dipping fault where the upper block of fault has moved downward in relation to the below block. Suggestively it is considered that rigid block rotation of southeastern Makran prompted this earthquake. The physical phenomena are commonly observed a prior strong EQs, the recent case studies have recommended geochemical and geophysical anomalies exciting before seismic event. These physical precursors are direct or indirect sign of commencement or movement of an irreversible rupture-producing physical process inside the preparation zone of an approaching earthquake or the manifestation of tectonic movement which takes place outside the preparation zone of an approaching event as an association with diastrophism in each distinct area former to the EQ. In this study, before the commencement of main shock, in the vicinity nearer to the epicenter of moderate or strong EQ tectonic stresses can emit Radon and greenhouse gases that initiate the variation in LST of the area usually due to the discharge of these gases to lower atmosphere [33]. Presence of p hole carriers in rocks can be another possible reason for LST anomalies associated with earthquake due to intensive tectonic stress applied to crustal rocks [43]. The presence of such carriers of electronic charges roots the higher infrared emission from the rock within a finite volume thus increasing LST [57]. However, post-EQ thermal anomalies are due to the failed attempt of normal condition re-establishment in effect of aftershocks even after the main event is over. Due to dominant residual stresses, the epicenter and close proximity undergo after-effects leading to the longer fading time for IR anomaly. The observation of common field for TIR anomaly of Awaran is a significant contribution.

The Kashmir EQ of shallow hypocentral depth 10 km struck the Pakistani region on September 24, 2019. The EQ occurred due to the movement of the reverse Jari Kass Fault in the Mirpur region, which has many active fault splays. A shallow blind fault line was outlined, fetching both the Jari Kass fault line and the Samwal fault on the surface exposition of this blind fault line (see, USGS webpage). Moreover, low magnitude EQs as aftershock series had started in the Mirpur region after restoring of the Mangla repository. Along these fault lines, there was a likelihood that the uprising may have enhanced the hydrostatic pressure that had been activating increment in the strain buildup. Another possibility that if the EQ event could be the result of the shallow reverse faulting close to the Indian and Eurasian plates. Focal mechanism provisions are not all around

compelled yet regularly show invert slip on a shallow or steeply plunging fault (e.g., USGS). Shallow east-west striking converse faulting is steady with the tectonics of the area. EQs and active faults in northern Pakistan and adjacent parts of India and Afghanistan are the immediate outcomes of this impact. The boundary additionally delivers the most elevated mountain tops in the world, including the Karakoram Himalayan, the Pamir, and the Hindu Kush Ranges. As the Indian plate moves toward the north, it is being subducted or pushed underneath the Eurasian plate. At the site of this event, the Indian plate converges with the Eurasian plate at a pace of 39 mm/year.

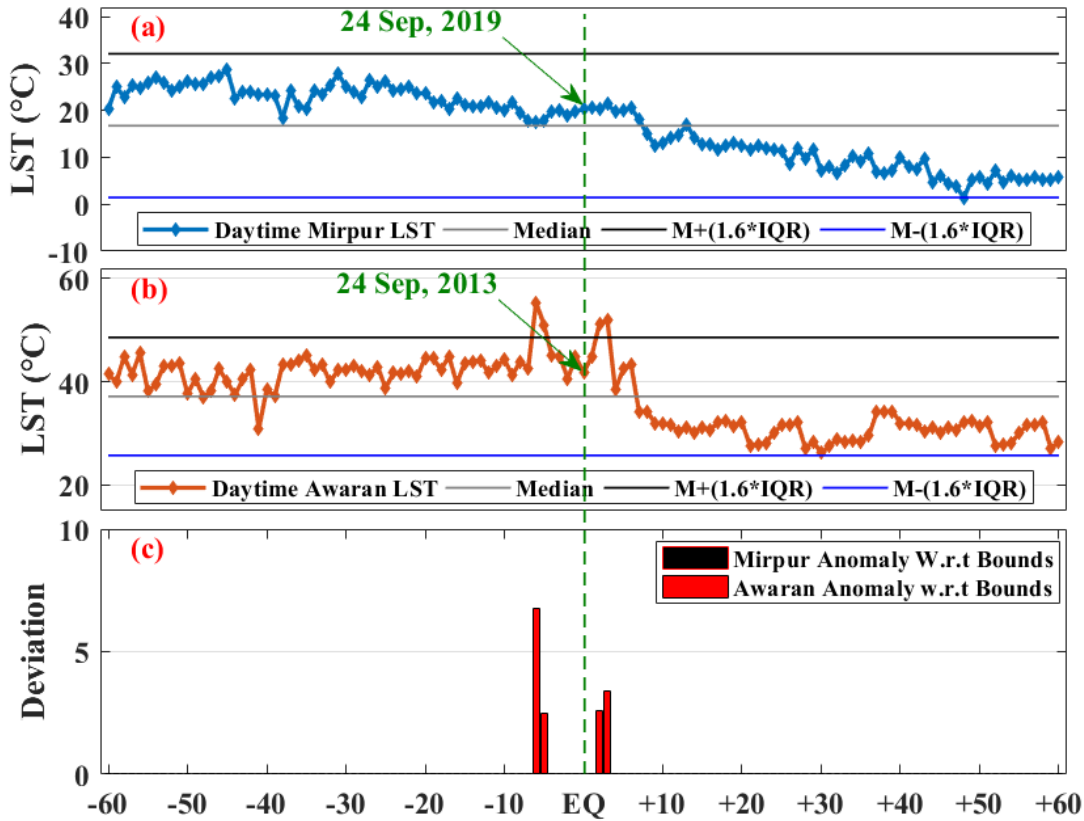
## **5.2 Results**

### **5.2.1 LST**

In Mirpur earthquake we observed that the daytime LST from MODIS shows no significant enhancement or depletion within the period of two months before and after the main shock which means we perceived a uniformity in land surface temperature throughout the 4 month time series, see [Figure 5.1a](#). Moreover, a low depletion far beyond the main shock after more than one month ( $LST < 1$ ) is observed but that has no relation with the EQ. One can see clearly no spatial enhancement in the analysis of LST daytime maps and a variation of less than 5°C occur over epicentral lineament regions. The spatial LST maps show quiet land surface temperature in daily MODIS images before and after the main shock. This can be due to the fact that intense radiative heating around the epicenter settles no significant variation in LST for daytime. As the daytime LST is more tightly coupled with the radiative and thermodynamic characteristics of the Earth's surface than standard air temperature measurements [51]. On the other hand, nighttime thermal variations after the main shock (probable post seismic thermal anomaly) is clearly visible beyond the bounds ([Fig 5.2a](#)). In this case the IQR upper bound was acquired to be 11.5 and lower bound -10.5. The anomalous pattern in night values is corresponding to 6 times more than the normal distribution of time series within 2 day after the main shock ([Fig. 5.2c](#)). The defined anomaly looks like the typical thermal anomaly at nighttime which is caused due to the vertical distribution of air temperature. Moreover, due to topography and cloud shadow regions the night time spatial imagery tends to avoid differential solar heating and the night-time data also suppresses the chances of partial sun illumination thus spatially pointing the thermal anomaly clearly [56]. In our study the spatial nighttime LST showed abnormal variations of more than 10.71 °C on the night immediately after the EQ, i.e., Sep 25, 2019. Furthermore, the significant rise in temperature as a result of abnormal emanation of

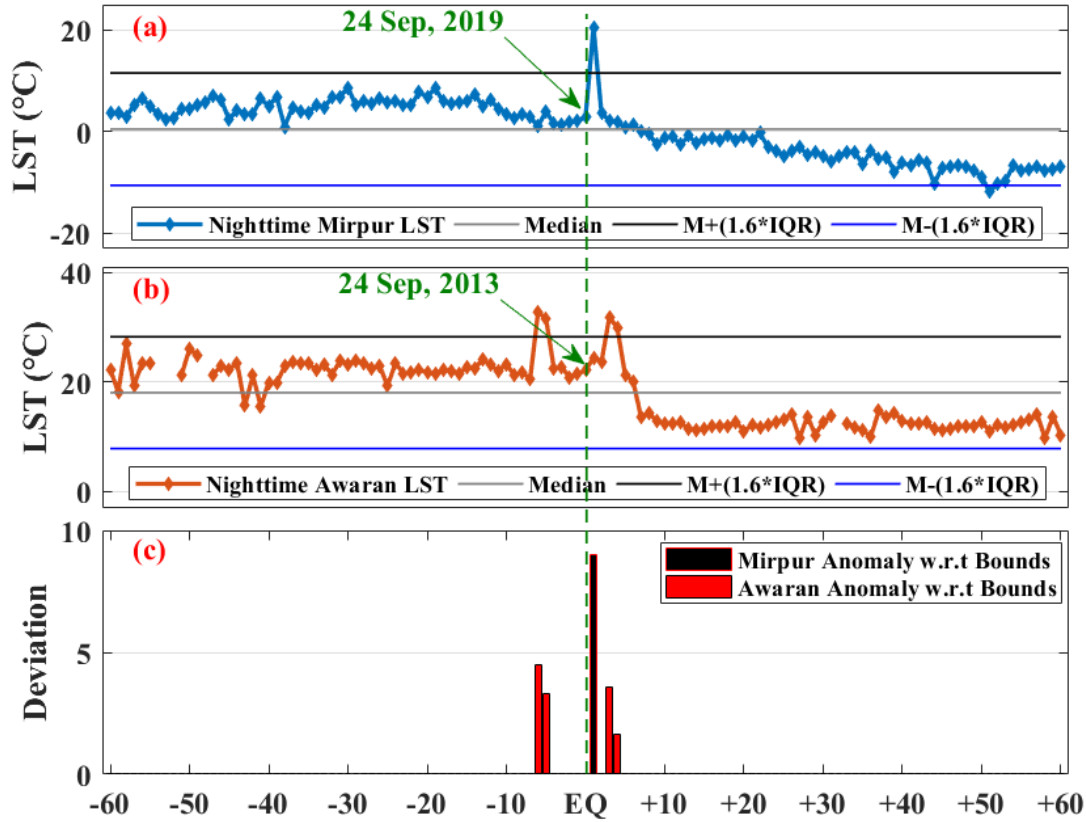
energy reached up to 20.6042 °C. It is showing that the obvious anomaly of 9.03 °C with respect to bounds is more than 19 % of the normal LST distribution in the deviation analysis.

While considering Awaran EQ, for daytime analysis of MODIS LST values, we recorded 2-6 °C rise in temperature, 5-7 days before the earthquake, and also there is a temperature rise of 2-3 °C within 5 days after the main shock as EQ was accompanied by a major aftershock referring to another possible EQ (Fig. 5.1b,c). Similarly, for nighttime analysis 3-4 °C abnormal temperature from normal distribution was observed within 5-7 days prior and similarly 1-3 °C increase around 5 days after the main shock (Fig 5.2b). Thus, an anomalous 2-6 °C and 3-4 °C for daytime and nighttime, respectively was acquired beyond the IQR bounds within 10-day window before and after the main shock event. Similar pattern of thermal anomaly was also observed by Shah et al. 2019 while conducting studies over Pakistan and Iran EQs (2010-2017) [67].



**Fig. 5.1.** Daytime MODIS LST averaged data time series with IQR bounds for both EQs, a) Daytime MODIS LST averaged data time series of Mirpur EQ with IQR bounds, b) Daytime MODIS LST averaged data time series of Awaran EQ with IQR bounds c)

The deviation of averaged LST values from IQR bounds where black bar shows Mirpur anomaly and red bar indicates Awaran anomaly and green line refers to EQ Day



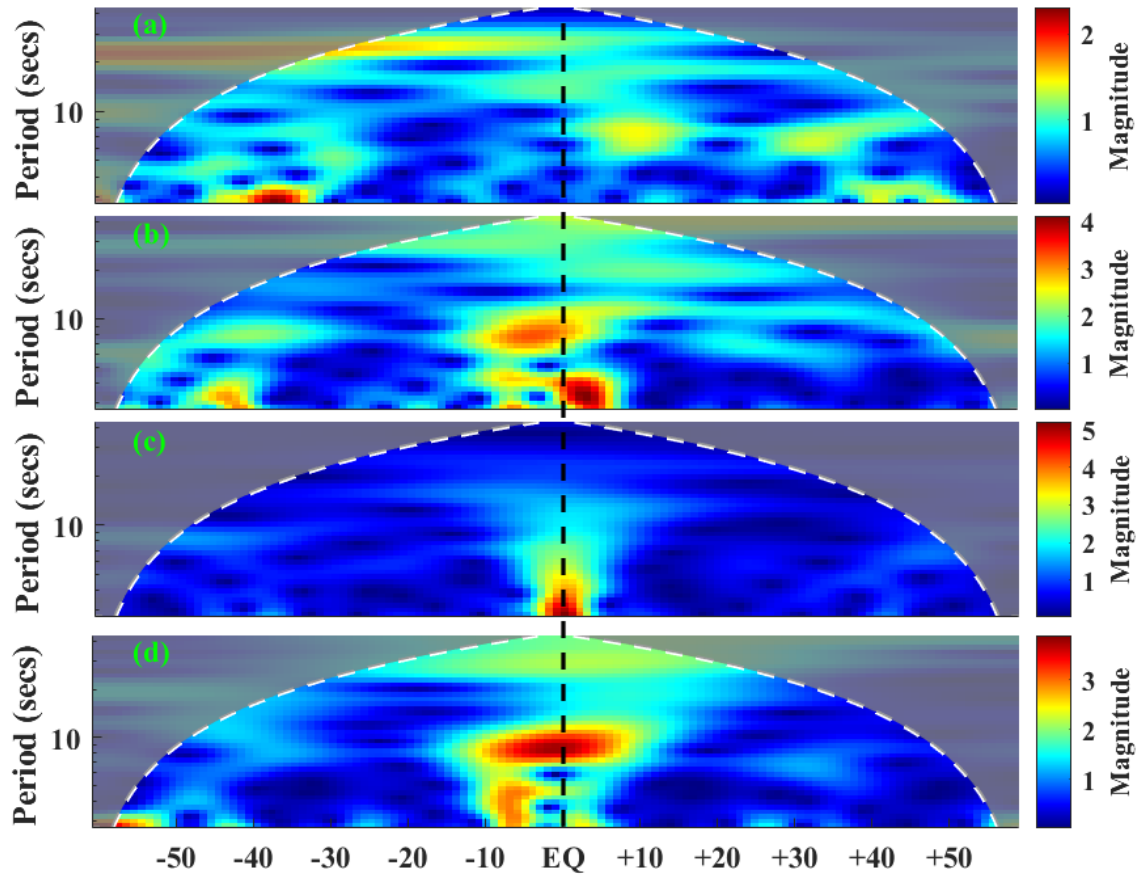
**Fig. 5.2.** Nighttime MODIS LST averaged data time series with IQR bounds for both EQs, a) Nighttime MODIS LST averaged data time series of Mirpur EQ with IQR bounds, b) Nighttime MODIS LST averaged data time series of Awaran EQ with IQR bounds c)

The deviation of averaged LST values from IQR bounds where black bar shows Mirpur anomaly and red bar indicates Awaran anomaly and green line refers to EQ Day

To detect the LST anomaly before the impending earthquake, the cwt was also applied. Including IQR, cwt based LST anomaly during daytime for Mirpur EQ 2019 also shows no significant enhancement consequent to the main shock of  $M_w$  5.6 EQ. The energy density spectrum shows

that 25 September 2019 manifested high values.

According to the energy density spectrum shows that 18 September 2013 is marked by high values and reveals enhancements with the continuous wavelet transform (cwt), obviously due to the impending EQ. The various wavelet functions and wavelet transformation levels over LST data for daytime and nighttime have been evaluated. Level 1 approximation information clearly showed the disturbing states that might be connected with an earthquake. For both events we observed significant results and [Figure 5.3](#) depict a clear anomaly within 10 day window before and after the main shock event for both time series, i.e., daytime and nighttime LST values.

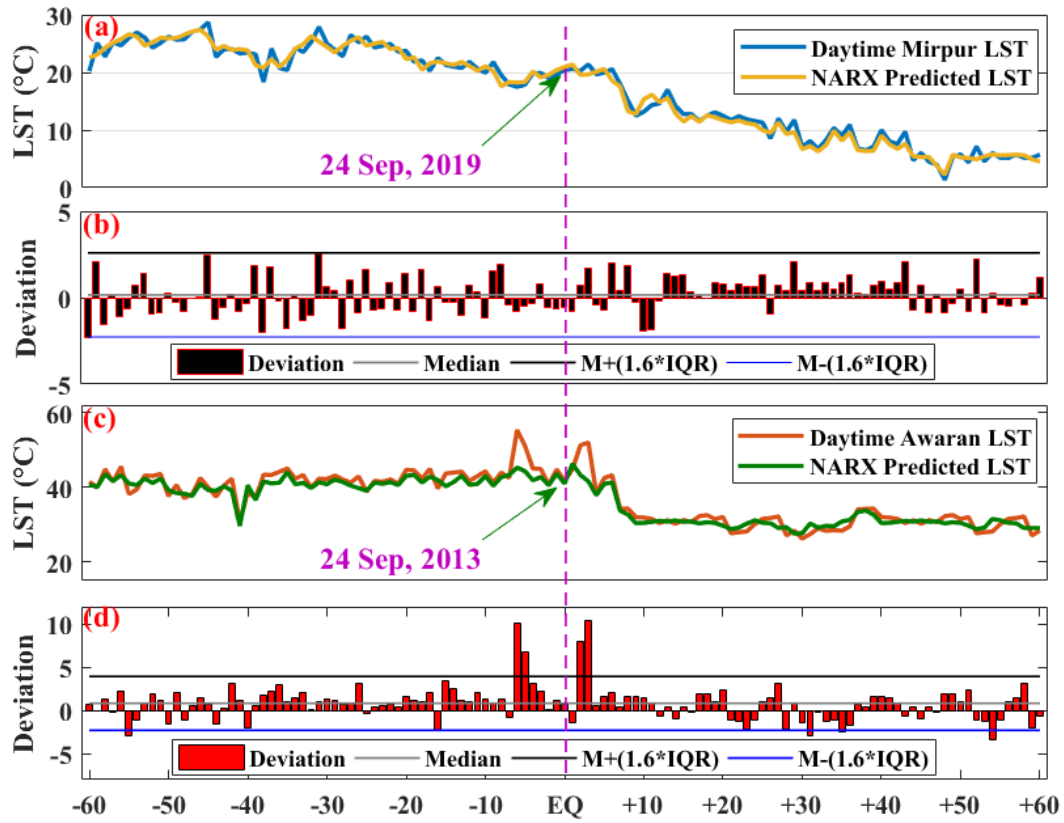


**Fig. 5.3.** Continuous wavelet transformation of LST time series of a) wavelet spectrum of Daytime LST of Mirpur EQ b) wavelet spectrum of Daytime LST of Awaran EQ c) wavelet



spectrum of Nighttime LST of Mirpur EQ d) wavelet spectrum of Nighttime LST of Awaran EQ

Additionally, the NN based LST anomaly during the day also showed no substantial variation for Mw 5.6 EQ (Mirpur, 2019). The observed and predicted LST from NARX have no clear EQ anomaly, which endorse the absences of enhancement in temporal LST (Fig 5.4a, b). While the spatial nighttime LST showed abnormal variations of more than  $10.71^{\circ}\text{C}$  on the night immediately after the EQ, i.e., Sep 25, 2019. Furthermore, the significant rise in temperature as a result of abnormal emanation of energy reached up to  $20.6042^{\circ}\text{C}$ . It is showing that the obvious anomaly of  $9.03^{\circ}\text{C}$  with respect to bounds is more than 19 % of the normal LST distribution in the deviation analysis (Fig. 5.5a). The time series data analyzed for nighttime showed that temperature predicted was  $4.8^{\circ}\text{C}$  after the EQ when trained by NARX technique, where the measured LST was  $20.6042^{\circ}\text{C}$ . Thus NARX predicted values are  $15.7^{\circ}\text{C}$  deviated from the observed values (Fig. 5.5b).

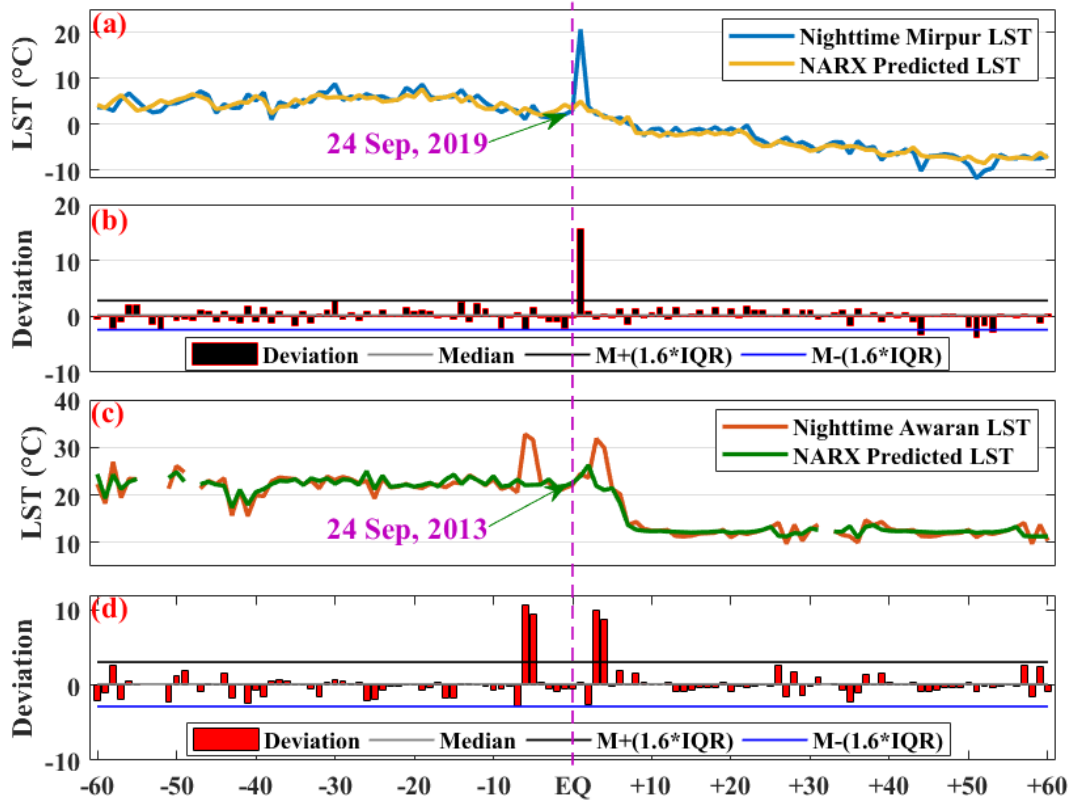


**Fig. 5.4.** Time series data variations between observed MODIS LST and predicted NARX

LST values for daytime with IQR bounds, a) Daytime Mirpur EQ NARX predicted and MODIS observed LST b) deviation between MODIS and NARX LST or Mirpur EQ values with confidence bounds, c) Daytime Awaran EQ NARX predicted and MODIS observed LST d) deviation between MODIS and NARX LST of Awaran EQ values with confidence bounds

The highest temperature observed for daytime series of Awaran EQ 2013 was around 55 °C and the temperature predicted by NARX was 45 °C by employing the previous inputs (Fig. 5.4c). Hence the deviation of about 10 °C was clearly detected before 7 days of the earthquake i.e., 18Sep and 7 °C deviation was recorded on 26<sup>th</sup> and 27<sup>th</sup> of September which is 2 to 3 days after the main shock as of 2 days before the major aftershock (Fig. 5.4d). The values exceeding the upper bound are called anomalies which in this case are falling within 10day window.

Considering the nighttime data series. We observed 10 °C deviation between MODIS observed LST and NARX predicted LST within 5-7 days before the EQ and after the main shock event difference of 9 °C was observed 2 days before the major aftershock (Fig. 5.5d).

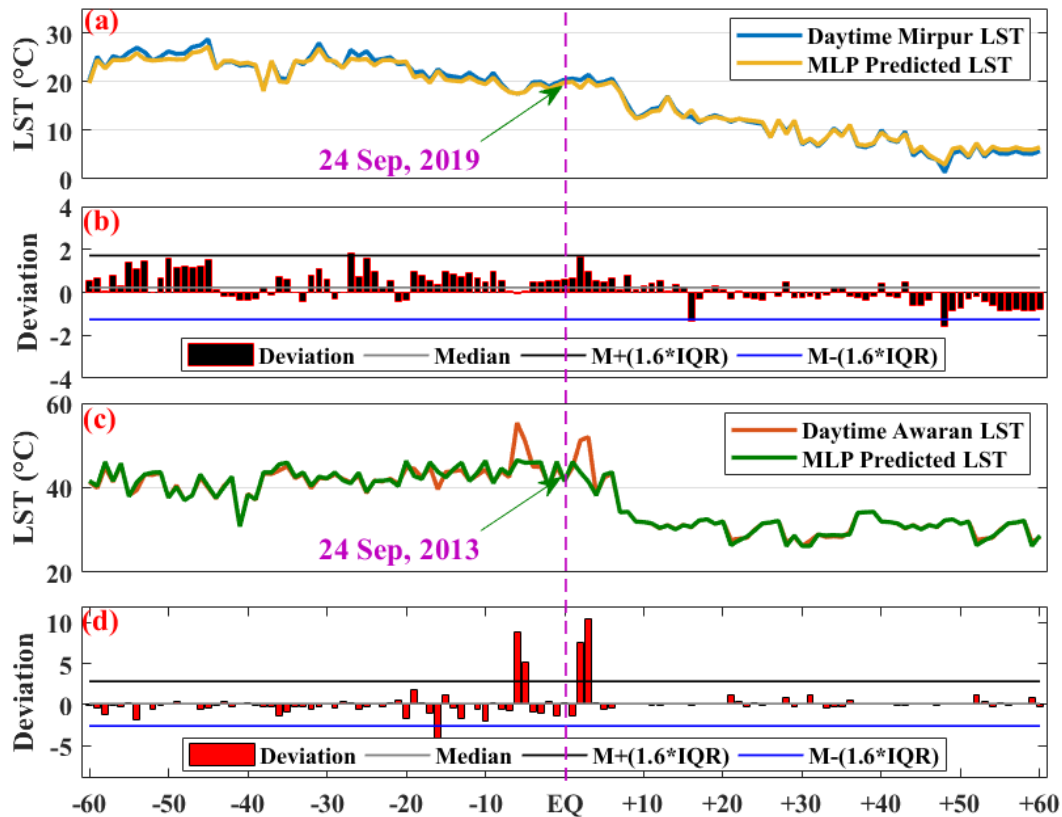


**Fig. 5.5.** Time series data variations between observed MODIS LST and predicted NARX LST values for nighttime with IQR bounds, a) Nighttime Mirpur EQ NARX predicted and MODIS observed LST b) deviation between MODIS and NARX LST or Mirpur EQ values with confidence bounds, c) Nighttime Awaran EQ NARX predicted and MODIS observed LST d) deviation between MODIS and NARX LST of Awaran EQ values with confidence bounds

By using another NN technique i.e., MLP we recorded significant results which are in

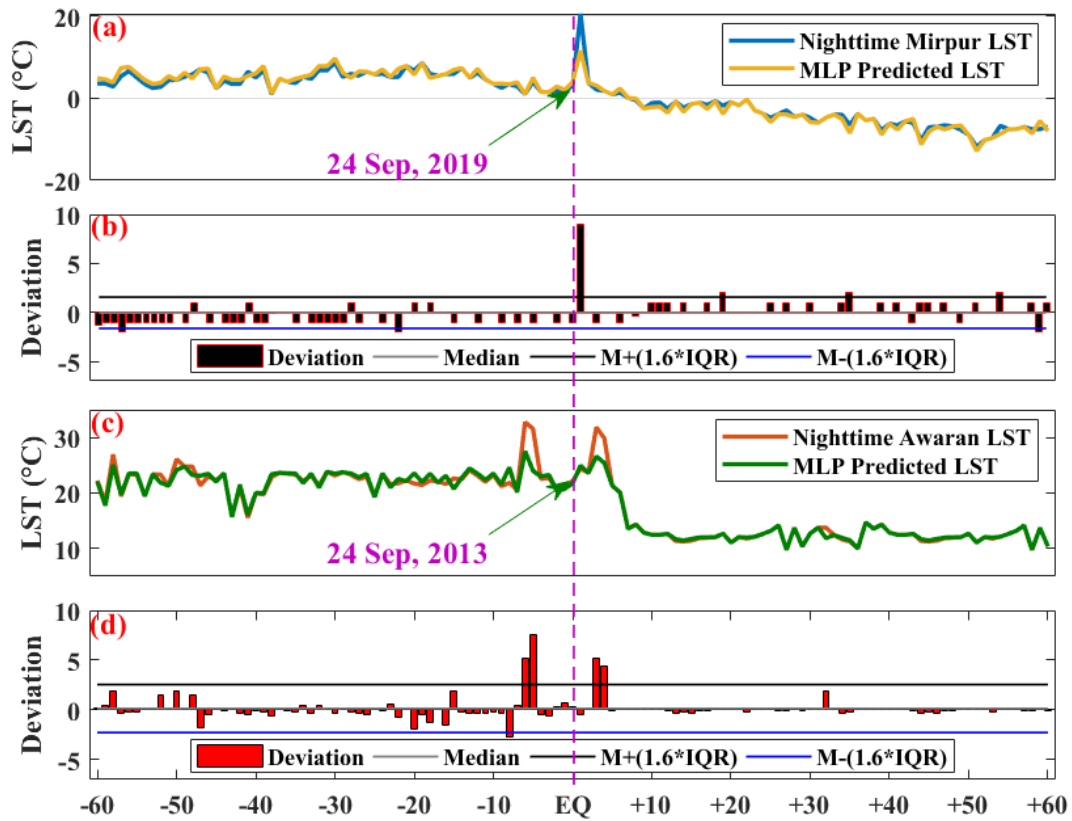
accordance with the previous techniques. We found that daytime LST recorded and predicted using MLP for Mirpur EQ does not provide any evidence of deviation or anomaly (Fig. 5.6a, b). When calculating for nighttime the MLP showed 11.6°C temperature on 25 Sep, 2019. MLP predicted temperature is 9°C deviated from observed values hence showing clear and definite anomaly (Fig. 5.7a, b).

Taking into consideration the case study of Awaran EQ the deviation in output predicted by MLP as compared to MODIS extracted LST is 9 °C for both day and night time data series during 5-7 days before EQ; whereas, post-seismic response has a 7-10°C variation within 2-4 days after the main shock, validating the post-earthquake response of a huge emanation of earthquake energy before the huge aftershock see Figures 5.6c, d and 5.7c, d. Therefore, the anomaly was gained by calculating deviation beyond the confidence bounds within 10 day window of main shock day.



**Fig. 5.6.** Time series data variations between observed MODIS LST and predicted MLP LST

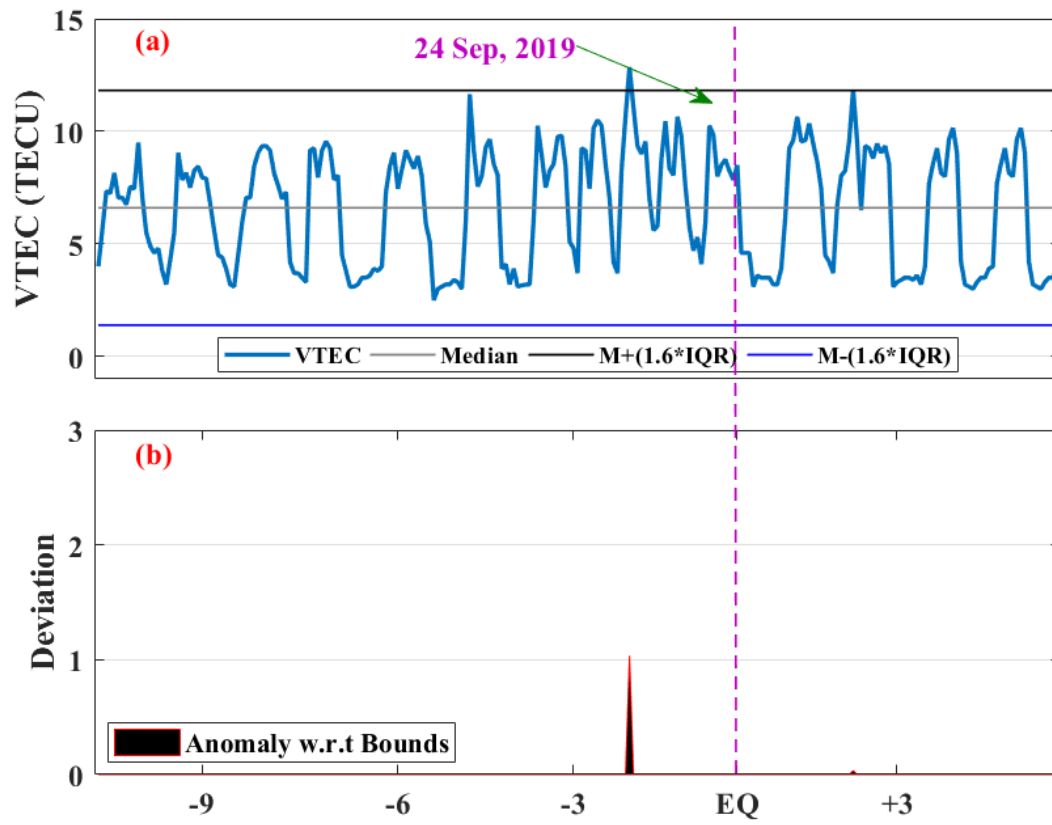
values for daytime with IQR bounds, a) Daytime Mirpur EQ MLP predicted and MODIS observed LST b) deviation between MODIS and MLP LST or Mirpur EQ values with confidence bounds, c) Daytime Awaran EQ MLP predicted and MODIS observed LST d) deviation between MODIS and MLP LST of Awaran EQ values with confidence bounds



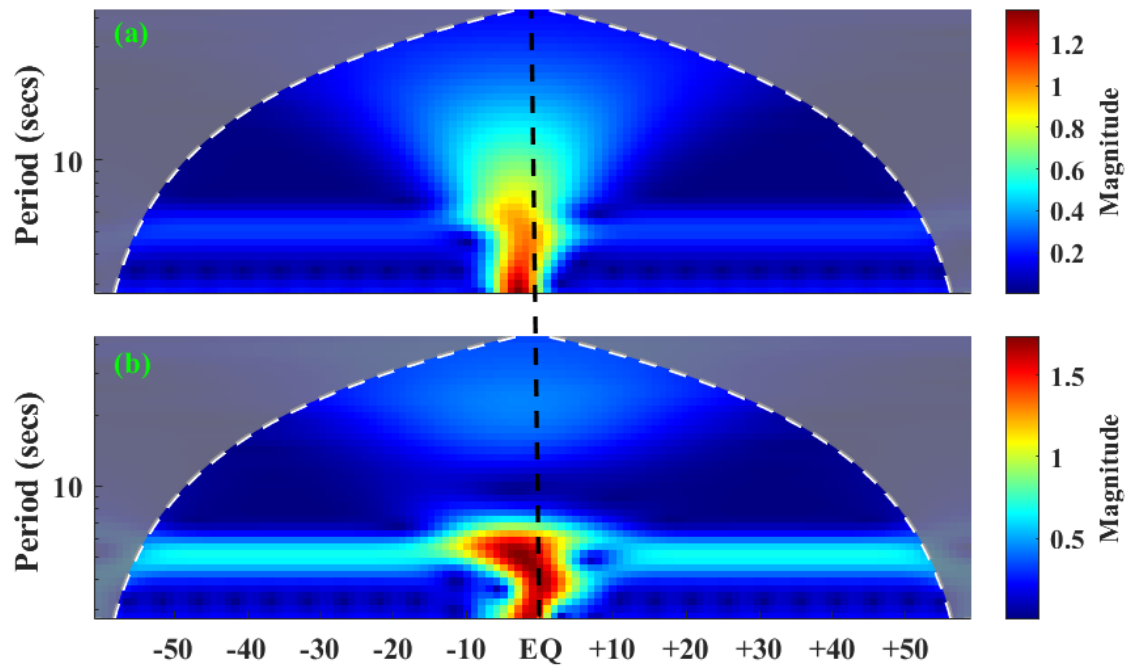
**Fig. 5.7.** Time series data variations between observed MODIS LST and predicted MLP LST values for nighttime with IQR bounds, a) Nighttime Mirpur EQ MLP predicted and MODIS observed LST b) deviation between MODIS and MLP LST or Mirpur EQ values with confidence bounds, c) Nighttime Awaran EQ MLP predicted and MODIS observed LST d) deviation between MODIS and MLP LST of Awaran EQ values with confidence bounds

### 5.2.2 TEC

The significant ionospheric TEC anomaly is observed on 22 September 2019 specifically during 7<sup>th</sup> hour exceeding the bounds (Fig 5.8). While no irregular value was observed after the main shock event i.e., no post EQ anomaly. The CWT identified the anomalous TEC value days in the form of spectrum where the magnitudes are depicted with scales within 2-3 days window see Fig 5.9a. The anomaly detected by implementing NARX was observed 6 days before till 3 days after the main event. On 19 September 2019 the predicted value deviated from observed value by 4.99TECU during 5<sup>th</sup> hour and the maximum deviation was observed of 5.23TECU during 15<sup>th</sup> hour of the same day. However on 20 September 2019 the few final hours of the day showed deviation around 3.6TECU and on 22 September 3.92TECU value deviated during 10<sup>th</sup> hour. While one day before the main event the initial hours of 23 Sep showed deviation around 3.5TECU (fig 5.10). On the day of main shock few hours before the EQ a noticeable deviation in TEC was observed (4.87TECU). After the EQ within 2-3 days an evident abnormality was also recorded using NARX technique. While evaluating MLP technique we observed an obvious deviation during 5<sup>th</sup> hour of 22 Sep 2019, the value deviated 7.14TECU where GPS value was 10.65TECU and MLP value was 3.50TECU see figure 5.11.

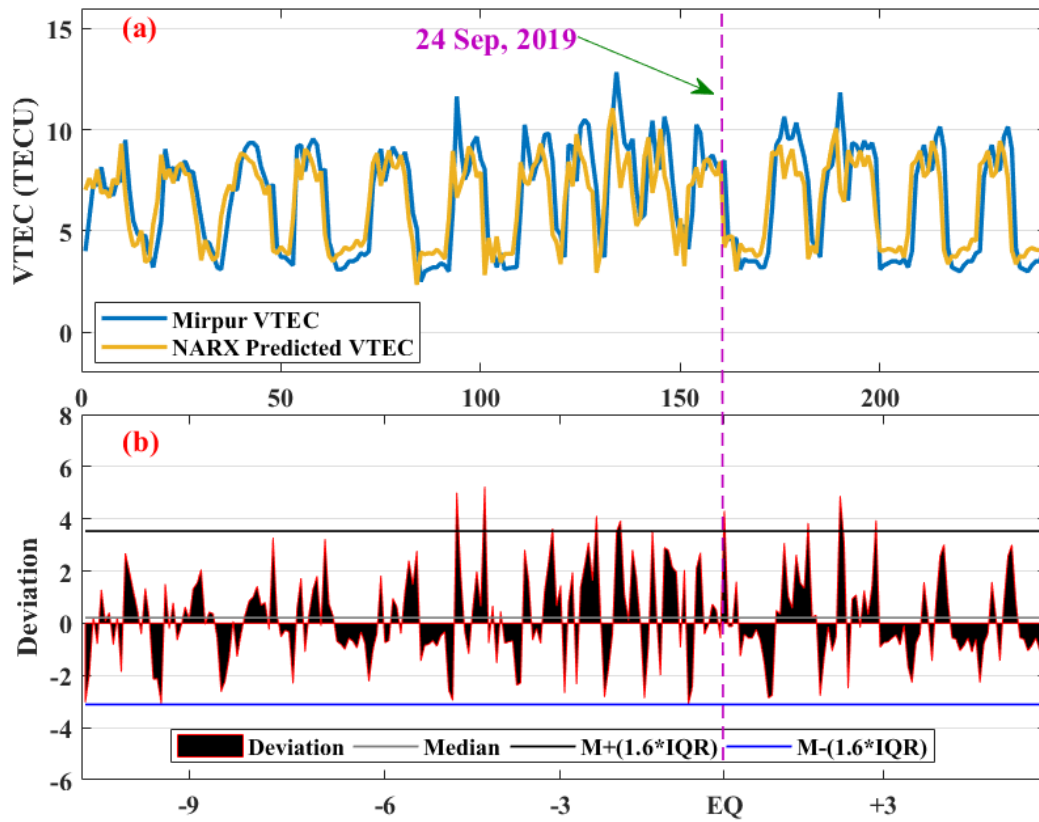


**Fig. 5.8.** a) Temporal TEC variations with upper and lower confidence bounds of Mirpur EQ  
b) values exceeding the bounds depicting the anomalous behavior



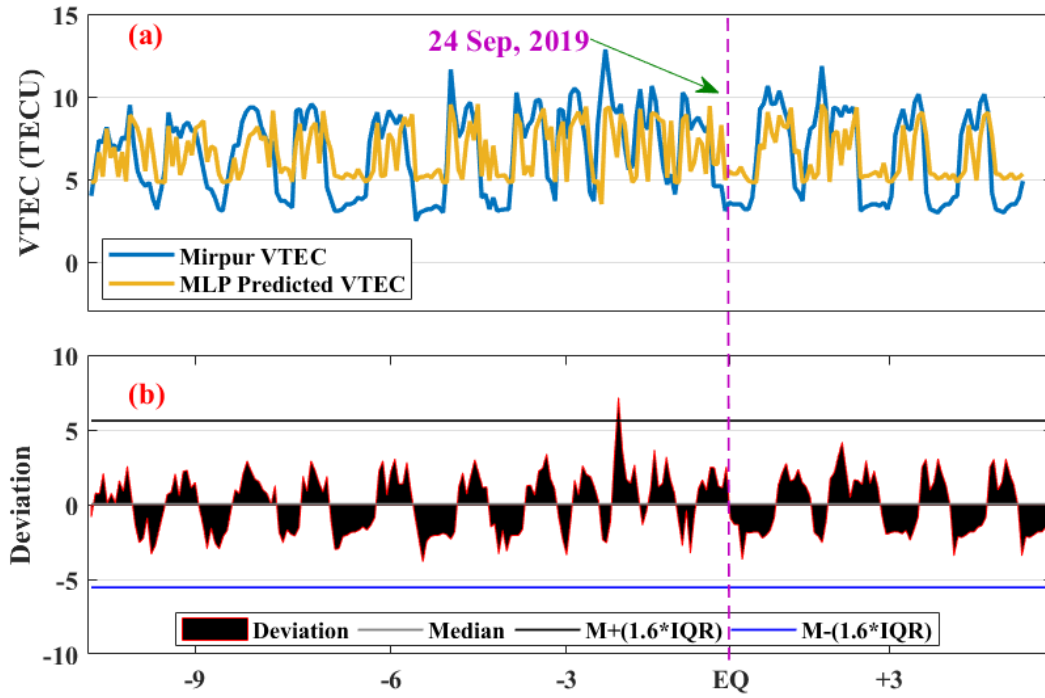
**Fig. 5.9.** a) shows wavelet spectrum for Mirpur EQ VTEC b) shows wavelet spectrum for Awaran EQ VTEC





**Fig. 5.10.** Comparison between GPS VTEC and NARX predicted VTEC of Mirpur EQ b)

values exceeding the bounds depicting the anomalous behavior



**Fig. 5.11.** a) Comparison between GPS VTEC and MLP predicted VTEC of Mirpur EQ b) values exceeding the bounds depicting the anomalous behavior

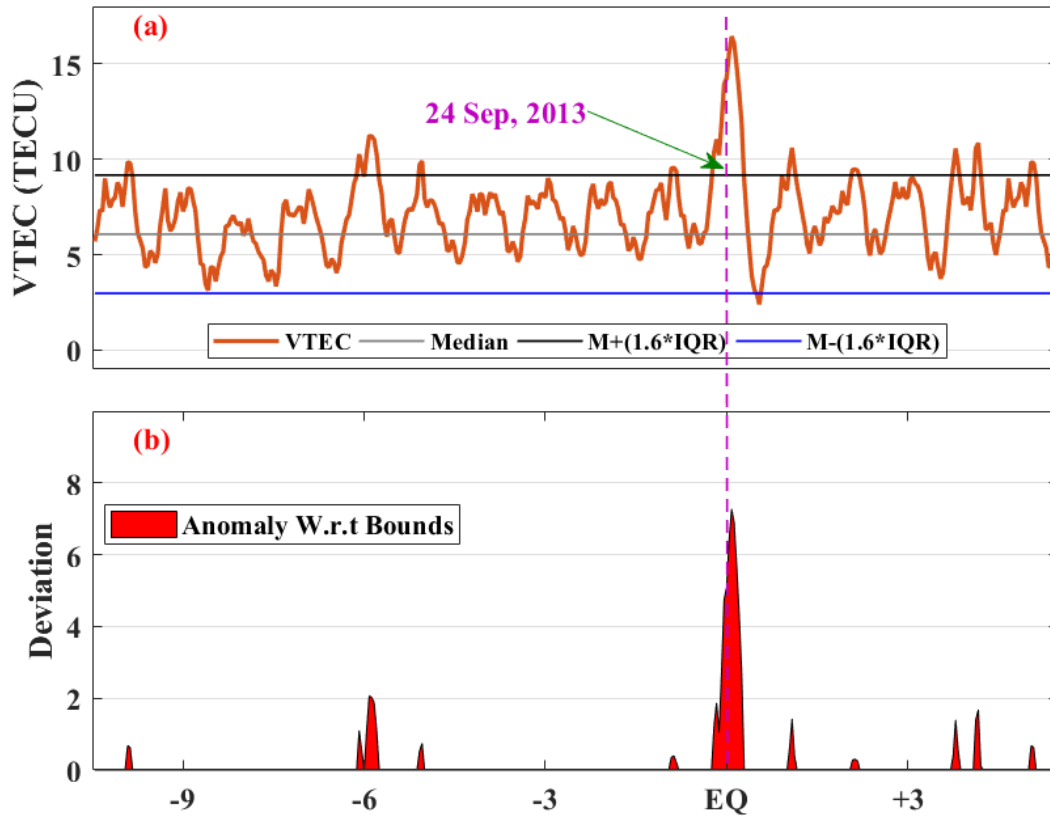
One can see clear ionospheric anomalies in the data of Sukkur GPS stations after one day of the main shock, which justifies the seismic energy emanation to ionosphere. The anomalous pattern was observed before the main event on 14 September 2013 which is 10 days before the EQ occurrence. While analyzing the temporal data of TEC we marked 0.67TECU rise in concentration on the 14<sup>th</sup> hours and 0.63C rise during 15<sup>th</sup> hour. Another upsurge of TEC was observed on 18 September i.e., 6 days prior to main event during 10<sup>th</sup>, 11<sup>th</sup>, and 13<sup>th</sup> till 17<sup>th</sup> hour ranging between 0.54TECU-2.06TECU. However, on 19 September 2013 the value by 0.34, 0.40 and 0.22TECU was greater than usual TEC values during 14<sup>th</sup>, 15<sup>th</sup> and 16<sup>th</sup> hour respectively. Moreover on the day of EQ occurrence we observed a series of disturbed values during 7<sup>th</sup> (1.21TECU), 8<sup>th</sup> (1.87TECU) 9<sup>th</sup> (1.04TECU) and 10<sup>th</sup> (2.73TECU), 11<sup>th</sup> (4.75TECU), 12<sup>th</sup> (5.07TECU), 13<sup>th</sup> (6.42TECU), 14<sup>th</sup> (7.27TECU), 15<sup>th</sup> (6.90TECU), 16<sup>th</sup> (5.69TECU) 17<sup>th</sup>

(4.33TECU) and finally on 18<sup>th</sup> (2.68TECU) hour. Ionospheric anomalies with several hours to several days before the main shock onset were widely reported [71][49][79], but interestingly we also logged few post seismic irregularities in TEC.

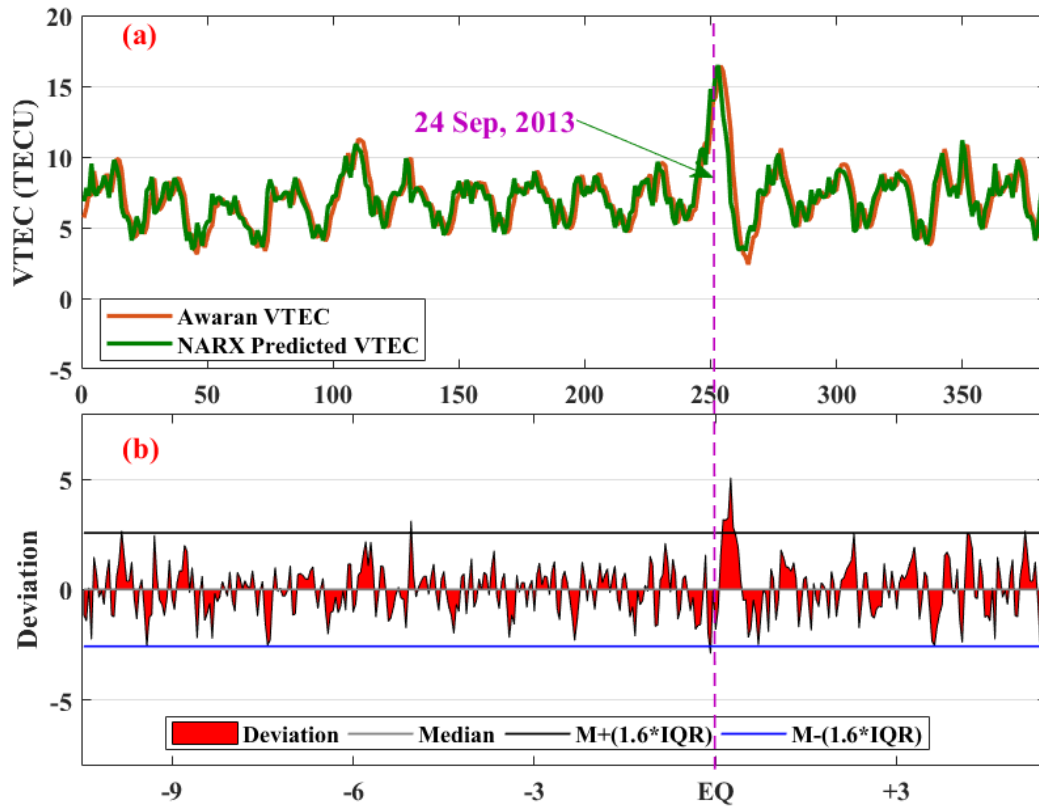
Nevertheless the observed irregularity TECU values after the earthquake was on 25 September 2013 during 13<sup>th</sup> 14<sup>th</sup> 15<sup>th</sup> hours consuming 0.58TECU, 1.42TECU and 0.38TECU deviation with respect to the remaining hours. Furthermore during 14<sup>th</sup> 15<sup>th</sup> and 16<sup>th</sup> hour of 26 September 2013, 0.25 to 0.30 deviation in TECU was noticeable. On 28 September 2013 according to USGS a major aftershock of Mw 6.4 also occurred which also showed abnormality in TEC values during frequent hours of the day e.g., 6<sup>th</sup> (0.38TECU), 7<sup>th</sup> (1.39TECU), 8<sup>th</sup> 0.55TECU and 15<sup>th</sup> (1.4TECU), 16<sup>th</sup> (1.68TECU) 17<sup>th</sup> (0.11TECU) hours. Moreover on 29 September 2013 two mid-day hours e.g. 13<sup>th</sup> and 14<sup>th</sup> also indicated deviation of 0.67TECU and 0.63TECU (Fig. 5.12).

While analyzing TEC data using cwt we observed the anomalous pattern of values around 6-7days prior to main event and within 3-4 days after the shock see figure 5.9b. While evaluating TEC data using Neural Networks we assessed the anomaly when the difference value ( $DX_i$ ) between the actual value ( $X_i$ ) and the predicted value ( $\widehat{X}_i$ ) exceeded outside the defined bounds.

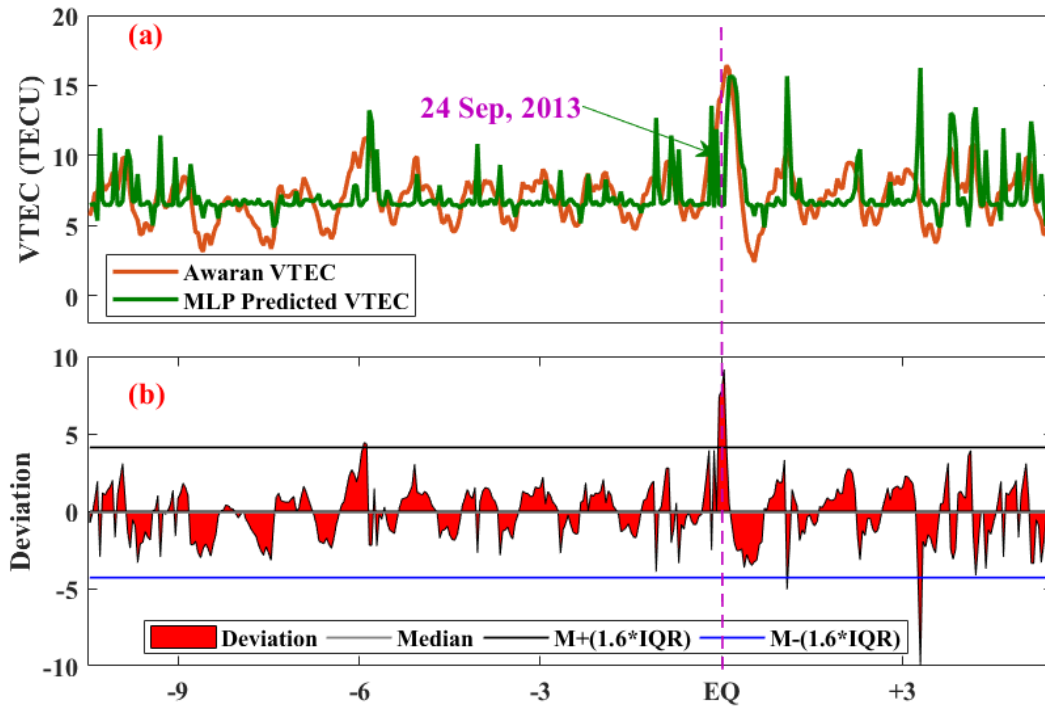
While implementing NARX model upper and lower bounds (2.58 and -2.6 respectively) were established. The pre earthquake anomaly of 3.13TECU was detected during 11<sup>th</sup> hour on 19 September 2013. During the midday to final hours (10<sup>th</sup> hour , 15<sup>th</sup>-20<sup>th</sup> hours) of the main event day i.e., 24 September 2013, the deviated values between observed and predicted values of TEC were clearly notable ranging from 2.47TECU to 5.09 TECU. As the record of post-EQ anomaly the NARX trained value is 2.57TECU deviated from observed TEC value during 15<sup>th</sup> hour on 29 September 2013 see figure 5.13. The bounds comprehended for MLP were 4.11 as upper bound and -4.29 as lower bound. On 18 September 2013 the deviation observed was 4.43TECU during 14<sup>th</sup> hour and 4.32TECU during 15<sup>th</sup> hour. However on the day of earthquake i.e., 24 September 2013 the values predicted were 7.41TECU deviated from the observed ones during 11<sup>th</sup> hour and 7.74TECU during 12<sup>th</sup> hour, the 13<sup>th</sup> hour recorded deviation of 9.13TECU. As an earthquake response the perturbed values of TEC were logged on 25 September 2013 during 14<sup>th</sup> hour (-5.06TECU). Another deviated value of -9.98TECU was predicted implementing MLP during 19<sup>th</sup> hour (Fig 5.14).



**Fig. 5.12.** a) Temporal TEC variations with upper and lower confidence bounds of Awaran EQ  
b) values exceeding the bounds depicting the anomalous behavior



**Fig. 5.13.** Comparison between GPS VTEC and NARX predicted VTEC of Awaran EQ b) values exceeding the bounds depicting the anomalous behavior



**Fig. 5.14.** a) Comparison between GPS VTEC and MLP predicted VTEC of Awaran EQ b) values exceeding the bounds depicting the anomalous behavior

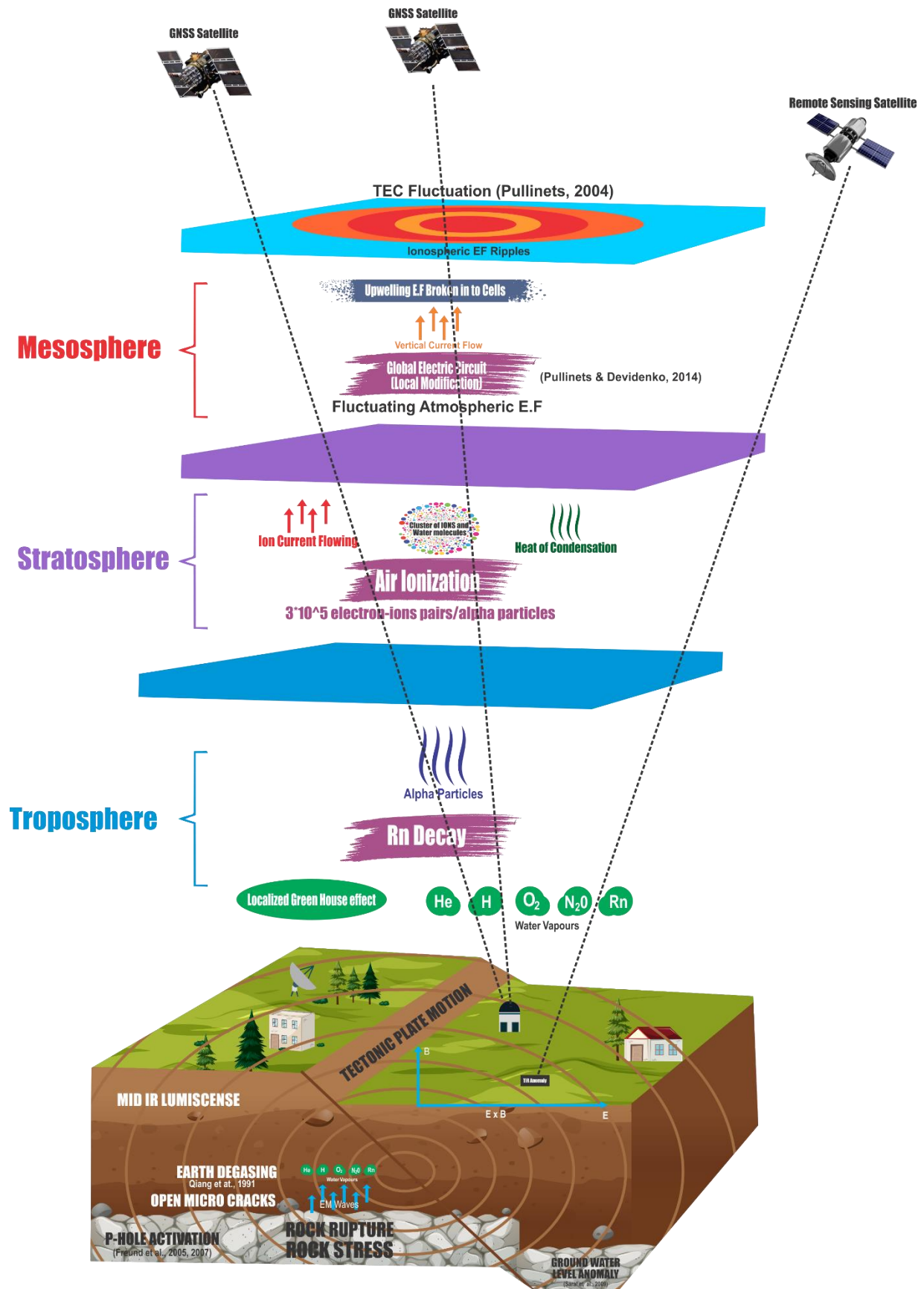
### 5.3 SUMMARY

This study focused on the coupling and evolution of seismo-ionospheric anomalies propagated through the lower atmosphere over the seismogenic zone. GPS based Earthquake (EQ) anomalies in the ionosphere and remote sensing based LST provide explicit evidences to study the coupling between seismic events, atmosphere, and ionosphere in epicentral breeding regions consequent to the EQ day in the preparation period. EQs are still not predicted, but the space-based EQ anomalies aid in the development of monitoring pre- and post-seismic precursors around the seismogenic zone and associated fault lineament regions. In this research work, tropospheric and ionospheric anomalies are investigated for EQs  $M_w < 5.4$  EQ from GPS based TEC and Land surface temperature. We noticed that atmospheric and ionospheric anomalies within 5–10 days before the main shock (EQ-precursors) and variations even after EQ (EQ response/ post-EQ). The enhancement and depletion in atmospheric and ionospheric anomalies occur due to the development of charges in squeezed rock within the vicinity of future EQ. The charge development

at the surface gives rise to a capacitor-like effect where the tectonic surface acts as one terminal and the ionosphere can be another side of a capacitor[86]. However, we emphasized in this study that LAIC coupling can be possible within 5-10 days prior the EQ or 5-10 days after the main shock (Fig. 5.15). The summarized behavior of seismo thermal and ionosphere anomalies see Table 5.1. Other than that low intensity anomalies in space measurements are false indicators in different data sets. It is noteworthy that spatial and temporal anomalies on LST are clear, which validates gas exhalation to complete the LAIC phenomena, which aid in the development of ionospheric anomalies at lower and upper edges of the ionosphere. The increase/decrease in ionospheric perturbations due to ions emanation over the epicenter due to the plasma flow direction. Results suggest that abrupt seismo TEC variations during EQ appeared as a result of the emanation of energy from the EQ prone region to the ionosphere. In conformity with previous studies and our conclusions, we believe that the seismo ionospheric and thermal anomalies from a number of ways can positively contribute in the prediction of EQs.

**Table 5.1.** Brief description of seismo thermal and ionosphere anomalies

| Methods                   | LST             |           |                    |                    | TEC                  |                        |                 |
|---------------------------|-----------------|-----------|--------------------|--------------------|----------------------|------------------------|-----------------|
|                           | Mirpur (Awaran) |           |                    |                    | Mirpur (Awaran)      |                        |                 |
|                           | Daytime         | Nighttime | Pre-EQ             | Post-EQ            | Daytime<br>Nighttime | Pre-EQ                 | Post-EQ         |
| IQR                       | ✗(✓)            | ✓(✓)      | Nil<br>(-7-to-5)   | 1 to 2<br>(2 to 4) | ✓(✓)                 | -2<br>(-6, -5 & EQ)    | Nil<br>(1 & 4)  |
| Wavelet<br>Transformation | ✗(✓)            | ✓(✓)      | Nil<br>(-10 to -5) | 1 to 2<br>(2 to 4) | ✓(✓)                 | -4 to -1<br>(-6 to EQ) | Nil<br>(1 to 2) |
| NARX                      | ✗(✓)            | ✓(✓)      | Nil<br>(-7 to -5)  | 1<br>(2 to 4)      | ✓(✓)                 | -5, -2<br>(-5 & EQ)    | 1 to 2<br>(Nil) |
| MLP                       | ✗(✓)            | ✓(✓)      | Nil<br>(-7 to -5)  | 1<br>(2 to 4)      | ✓(✓)                 | -2<br>(-6 & EQ)        | Nil<br>(1 & 3)  |





**Fig. 5.15.** Proposed LAIC Model

## **6 CONCLUSION AND FUTURE ASPECT**

In this study, standard, classical and intelligent algorithms were implemented for the detection of irregular variations in MODIS based LST and GPS based VTEC about the time of the Awaran earthquake and Mirpur EQs. Every three methods detect very striking irregular variations within 10 days prior to the earthquakes and also these events were followed by aftershocks referring to post EQ anomalies with 3-4 day window over the epicentres for the completion of LAIC mechanism. This study purposely portrayed and has vindicated the connotation of pre- and post-seismic LST and TEC anomalies with different EQs of different magnitude statistically, classical and by NN methods. Our research enthralled the local change in near-earth plasma caused by abnormal electric fields penetrating from the ground. An Earth's LAIC phenomenon was hypothesized by studying the synergy of numerous processes that may occur during a seismic event. Radon emission plays a crucial role in triggering chemical and physical reactions in the atmosphere, which alters ionospheric electron concentration over time. These atmospheric circulations cause the formation of large ion clusters, which alter the conductivity of the atmospheric boundary layer. As a result of this effect, plasma ionospheric concentration varies further. TEC and LST anomalies are more intense before shallow depth and larger than post-seismic ones around the pre-seismic durations. Since the LST anomalies occurred before the EQs, suggests that it may be a reliable indicator of an impending EQ, whereas the TEC anomalies might be caused by a driving force generated around the epicenter during EQ preparation. A large amount of energy is produced around the epicenter, implying the presence of an energy phenomenon that needs more investigation in the future. In most earthquakes, aftershocks occur because the rocks that formed the earthquake fault have moved to new positions and applied stress to neighboring faults. Aftershocks are generated as other faults release the new stress. However, this phenomenon has yet to be discovered and deserves a debate to find out how it works. Using multiple satellite observations, these anomalies after EQ contribute to developing LAIC hypothesis for seismic precursors. In order to enhance our knowledge of EQ precursors, more analyses with different data types and studies are still needed. It is also proposed that earthquake prediction will no longer be

a dream by expanding the current cluster of multi-parameter and interdisciplinary precursory techniques, including TEC, radon gas emanation, and other satellite data to remote sensing.

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